INTRODUCTION TO MANAGEMENT

What is Management?

Management is a form of work that involves co-ordinating an organisation's resources - land, labour, and capital. Management is the process of working with people and resources to accomplish organisational goals. Good managers do those things both effectively and efficiently. To be ***effective*** is to achieve organisational goals. To be ***efficient*** is to achieve goals with minimum waste of resources, that is, to make the best possible use of money, time, materials, and people. Some managers fail on both criteria, or focus on one at the expense of another. The best managers maintain a clear focus on both effectiveness and efficiency.

Managers seek to maximise the potential of their people and co-ordinate their efforts toward achievement of some predetermined goal. All organisations establish a variety of goals to be accomplished through the efficient utilisation of human and nonhuman resources. Management must perform at maximum efficiency in order to achieve organisational goals and meet the expectations of the many internal and external groups that are affected by, and exert influence upon, the organisation.

The Internal and External Environment

The internal environment of an organisation consists of the conditions and forces within an organisation e.g. owner, employees, physical environment, board of directors, culture, etc. The external environment is everything outside an organisation's boundaries that might affect it e.g. competitors, customers, suppliers, strategic allies, regulators, socio-cultural dimension, economic dimension, technological dimension, international dimension, political-legal dimension.

Levels of Management

Three levels of management exist. ***Top management*** establishes the goals of the organisation and the actions necessary to achieve these goals. ***Middle management*** develops department goals and actions necessary to achieve the organisational goals. The ***supervisor*** manages operative employees.

The organisational level occupied by the manager, which may be top, middle, or supervisory, has a direct impact upon the managerial skills and abilities required of the manager and the types of roles he or she will be expected to play. Each manager tends to develop a "style" or approach to the job which reflects his or her personality, the type of organisation, and the relationships with superiors, subordinates, and others.

Top-level Managers

Top-level Managers are the senior executives of an organisation and are responsible for its overall management. Typically top-level managers, often referred to as strategic managers, focus on long-term issues and emphasise the survival, growth, and overall effectiveness of the organisation.

Top managers are concerned not only with the organisation as a whole, but also with the interaction between the organisation and its external environment. This interaction often requires managers to work extensively with outside individuals and organisations.

The chief executive officer (CEO) is one of top-level manager found in large companies or (corporations). This individual is the primary strategic manager of the firm. Others include the chief operating officer (COO), company presidents, vice presidents, and members of the top executive committee.

Middle-level Managers

Middle-level managers are located in the organisation's hierarchy between top-level management and the supervisory management (or frontline managers). Sometimes called ***tactical managers***, they are responsible for translating the general goals and plans developed by strategic managers into more specific objectives and activities.

Traditionally, the role of the middle manager is to be an administrative controller who bridges the gap between higher and lower levels. Middle-level managers take corporate objectives and break them down into business unit targets; put together separate business unit plans from the units below them for higher-level corporate review; and serve as linchpins of internal communication, interpreting and broadcasting top management' priority downward, and channelling and translating information from the frontlines, upward.

The evolving role of middle managers now requires them to be not only administrative controllers but also development coaches to the people who report to them. They should support the activities of their people, and coach people to become more entrepreneurial and innovative. They should help develop the people to develop the business.

Middle managers should ensure that those reporting to them keep long-term strategic objectives and short-term, more immediate operating priorities in balance. For example, this may involve forcing a frontline manager to make major cuts in his unit in order to meet financial objectives while at the same time fighting against attempts to close a unit down.

Supervisory or Frontline Managers

Supervisory managers are sometimes referred to as ***operational managers***. They are lower-level managers who supervise the operations of the organisation. These managers often have titles such as Supervisor or Foreman. They are directly involved with non-management employees, implementing specific plans developed with middle managers. This role is critical in the organisation, because operational managers are the link between management and non-management personnel.

Traditionally, supervisory or frontline managers have been directed and controlled from above, to make sure that they successfully implement operations in support of company strategy. But in leading companies the role has expanded. Whereas the operational execution aspect of the role remains vital, in leading companies frontline managers are increasingly called upon to be innovative and entrepreneurial, managing for growth and new business development. Frontline managers are not only allowed to initiate new activities, but are expected to by their top and middle level managers.

Why study Management?

- **Managers are Universal**

Managers work in all types of organisations, at all levels, and in all functional areas. Large and small businesses, hospitals, schools, governments, and churches benefit from efficient and effective management. The leaders of these organisations may be called executives, administrators, principals, or pastors, but they are all managers and are responsible for the success or failure of the organisation. This success or failure is reflected in a manager's career. For example, when a CEO saves a failing company, the board rewards this success with bonuses. When a professional football team starts losing, the owner fires the coach, not the team.

Managers are also found in each functional area of an organisation. Accountants are promoted to accounting department heads and project team leaders, sales representatives become sales managers, writers become editors, and nurses become nursing administrators. Management skills are important to anyone who intends to pursue a career. No matter where one intends to work, effective managers are a necessity.

- **Management by Common Sense**

Managers should plan for the future; organisations should adapt to their environments; and managers should identify things their employees value and offer them as rewards for high performance. Most of this sounds like common sense.

Although some of the management concepts may sound like common sense, developing them on your own or putting them into practice in the proper way, at the right time, and under appropriate conditions is difficult. If management was all common sense, there would not be so many failing companies, and so many people who complain about their incompetent bosses. Our natural ability to make the right decision or take the proper action – our common sense – can be improved greatly through systematic study.

- **Management by Experience**

A management course can offer valuable preparation and supplement one's experience. The concepts from a management course can help one make sense out of one's experiences more quickly. These concepts will provide one with a head start in one's management career. Furthermore, the concepts from a management course are an organised summary of the experiences of many managers.

Entrepreneurship and Management

A basic distinction is often made between a manager and an entrepreneur. An **entrepreneur** conceives the idea of what product or service to produce, starts the organisation, and builds it to the point where additional people are needed. At this point, the entrepreneur can either make the transformation to professional manager or can hire one. A professional manager performs the basic management functions for the ongoing organisation. Note, however, that an entrepreneur must perform many if not all of the basic management functions in starting and building an organisation.

Approaches to Analysing Management

Several approaches have been used to analyse the job of management. One common approach examines management by categorising the functions (work) performed by managers. A second approach looks at the roles that managers perform. A third approach analyses the skills required of managers.

The Functions of Management

What can managers do to be effective and efficient? The management process, properly executed, involves a wide variety of activities, including ***planning***, ***organising***, ***staffing***, ***leading*** and ***controlling***. These basic activities are the traditional functions of management.

- **Planning** involves deciding what objectives to pursue during a future period and what to do to achieve those objectives.
- **Organising** involves groping activities, assigning activities, and providing the authority necessary to carry out the activities.
- **Staffing** involves determining human resource needs and recruiting, selecting, training, and developing human resources.
- **Leading** involves directing and channelling human behaviour toward the accomplishment of objectives.
- **Controlling** involves measuring performance against objectives, determining causes of deviations, and taking corrective action where necessary.

Management Roles

A role is defined as an organised set of behaviours belonging to an identifiable job. The roles of a manager can be grouped into three major categories: interpersonal, informational, and decisional. Formal authority and status together generate certain interpersonal roles. The interpersonal roles in turn determine the informational roles of the manager. Finally, access to information, authority and status place the manager at a central point in the organisational decision making process.

Management Skills

Skills are specific abilities that result from knowledge, information, practice, and aptitude. The three skills required in management are technical skills, conceptual and decision skills, and interpersonal and communication skills. When the key management functions are performed by managers who have these critical management skills, the result is a high-performance work environment.

- **Technical Skills**

A **technical skill** is the ability to perform a specialised task that involves a certain method or process. Most people develop a set of technical skills to complete the activities that are part of their daily lives. When you leave school, you will have a set of technical skills that will provide you with the opportunity to get entry-level position. Accounting majors will develop many of the basic skills needed to conduct audit. Information systems majors will have important computer skills. Marketing majors may know pricing, market research, and sales techniques.

Managers may rely less on their basic technical skills as they rise through an organisation, but these skills give them the background for their new responsibilities, as well as an appreciation of the activities of others in the firm.

- **Conceptual and Decision Skills**

Conceptual and decision skills involve the manager's ability to recognise complex and dynamic issues, to examine the numerous and conflicting factors that influence these issues or problems, and to resolve the problems for the benefit of the organisation everyone concerned.

Managers use conceptual and decision skills when they consider the overall objectives and strategy of the firm, the interaction among different parts of the organisation, and the role of the business in its external environment.

- **Interpersonal and Communication Skills**

Interpersonal and communication skills influence the manager's ability to work well with people. These skills are often *people skills*. Managers spend the great majority of their time interacting with people. Because managers must deal with others, they must develop their abilities to lead, motivate, and communicate effectively with those around them. The ability to get along with many diverse types of people and exchange information with them is vital for a successful management career.

While there are still plenty of traditional managers around, concentrating on being the boss, giving orders, and carefully monitoring employees, many believe that the manager of today and of the future must focus on interpersonal skills such as being a team player, sharing information with others, and teaching and helping people learn.

Schools of Management Thought

The study of management is, at best an inexact science. Management thought is generally divided into classical, behavioural, and quantitative schools.

The Classical Approach

A. *Early management theorists* focused on the search for scientific methodologies capable of enhancing worker productivity and later broadened that focus to include the study of management functions.

1. Robert Owen was a social reformer who blended a search for scientific management principles with genuine concern for the worker.
2. Charles Babbage advocated many of the techniques of scientific management that were later popularised by Taylor, Gantt, the Gilbreths and others.

B. *Scientific management theorists* employed a systematic, analytical study of work to identify the most efficient method for performing a particular task and to train workers in that method.

1. Frederick Winslow Taylor, the "father of scientific management," focused his efforts upon the identification of the one best way to perform a task through experimentation in the workshop.
2. Henry L. Gantt, the first to suggest a task-and-bonus plan for both supervisors and workers, is best known for the development of a bar chart for planning and controlling work activities. This chart was the precursor of today's programme evaluation and review technique (PERT).
3. The Gilbreths pioneered many of the techniques and devices utilised in time and motion study, suggested a merit rating system, and began the research that eventually led to modern research on ergonomics.
4. Harrington Emerson, who coined the term "efficiency engineering," developed a system of twelve principles of efficiency for the railroad industry that could form a basis for building a sound management system even today.

C. *Classical administrative and organisation theorists* sought to assist managers above the supervisor level by studying administrative practices, organisational structure, the functioning of formal and informal organisations, and the social environment of management.

1. Henri Fayol, perhaps the single greatest contributor to the field of classical administrative theory, defined five managerial functions and developed fourteen seminal principles of management.

a. *Division of work* builds upon specialisation to allow workers to build upon the expertise of other workers in a logical sequence.
b. *Authority and responsibility* should be commensurate.
c. *Discipline* requires adherence to all plans that govern business operations.
d. *Unity of command* dictates that an employee should receive directives from only one superior.
e. *Unity of direction* requires that all activities geared toward achieving the same objective be directed and controlled.
f. *Subordination of individual to general interest* requires all organisation members to place interests of the organisation above their own.
g. *Remuneration of personnel* requires fair and equitable wages.
h. *Centralisation* refers to the appropriate concentration or dispersion of authority in the organisation.
i. *Scalar chain* requires all organisational requests and directives to follow the formal chain of command.
j. *Order* insists upon the maintenance of logical and functional organisational relationships.
k. *Equity* emphasises fairness and kindness in employee relations.
l. *Stability of tenure of personnel* is essential to the maintenance of a productive work force.
m. *Initiative,* the ability to think a problem through and develop a plan of action, must be developed in all subordinates.
n. *Esprit de corps*, or harmonious unity, is essential to organisational success.

2. Chester I. Bernard, the originator of the management systems concept is best known for his analysis of how formal and informal organisations operate and for his "acceptance" theory of authority.

a. Formal organisations are created when several people agree to work together to accomplish goals that they could not achieve separately.
b. Subordinates will accept and submit to authority if they understand the directive, feel the directive is consistent with the organisation's purpose, see no conflict with their personal beliefs, and are able to perform the task as directed.

3. Mary Parker Follet bridged the gap between the classical and behavioural schools of management by stressing the importance of harmony and cooperation among group members.

a. Effective leadership depended on the consent of the group.
b. The right to exercise authority depended on the situation; it was not vested in an organisational position.

D. *Other contributors to the classical school* expanded upon and enriched the principles developed by the seminal management theorists.

1. Alexander Church conceptualised the functions o managers and demonstrated their interrelationships.
2. Oliver Sheldon was perhaps the first to recognise management's social responsibility.
3. James Mooney elevated the scalar the scalar and functional processes to the status of "pillars" of organisational theory.

4. Max Weber contributed to the understanding of the bureaucratic organisation.
5. Lyndall Urwick synthesised the philosophies of other contributors.

The Behavioural School

A. The behavioural school of management emphasises the interactions of people in an organisation.
B. *Elton Mayo* and the *Hawthorne experiments*, conducted at Western Electric's Hawthorne, Illinois, plant demonstrated the importance of sociological and psychological factors in productivity.

1. The Hawthorne *effect* demonstrated that workers were more productive when their efforts were noticed.
2. The Hawthorne studies became a major factor in the growth of the human relations school of management.

C. Criticism of the Hawthorne experiments focused on the methods employed and suggested that incentive pay and better working conditions, not simple observation, accounted for improved productivity.
D. Other Hawthorne experiments failed to replicate the Hawthorne effect.
E. Other behaviourists have made contributions to the study of leadership styles, motivational techniques, behaviour modification approaches and a variety of other human and organisational behavioural concepts.

The Quantitative School

A. The quantitative school of management uses sophisticated mathematical models to simulate business problems.
B. Advances in computer technology have facilitated the use of linear programming, network analysis, queuing theory, and other sophisticated techniques.
C. The use of mathematical models allows the analyst to manipulate problem-related data, which must be accurate, to identify the optimal solution.

More Recent Theories

A. ***Contingency theory*** is based on the premise that situations dictate managerial action - different situations call for different approaches.
B. **The *systems theory*** holds that a manager must be able to see the interrelationships of the different parts of his or her organisation and must understand how the organisation fits into its larger environment.
C. ***Theory Z***, as the Japanese approach to management is called, emphasises workers as the key to increased productivity and economic growth and incorporates traditional Japanese collective decision making and control mechanisms.

MANAGERIAL PLANNING

What is planning?

Planning is the process of deciding what objectives to pursue during a future time period and what to do to achieve those objectives. It is the primary management function and is inherent in everything a manager does.

Why Plan?

It is futile for a manager to attempt to perform the other management functions without having a plan. Managers who attempt to organise without a plan will find themselves reorganising on a regular basis. The manager who attempts to staff without a plan will be constantly hiring and firing employees. Motivation is almost impossible in an organisation with continuous reorganization and high employee turnover.

Planning enables a manager or organisation to affect rather than accept the future. By setting objectives and charting a course of action, the organisation commits itself to "making it happen." This allows the organisation to affect the future. Without a planned course of action, the organisation is much more likely to sit back, let things happen, and then react to these happenings in a crisis mode.

Planning provides a means for actively involving personnel from all areas of the organisation in the management of the organisation. Involvement produces a multitude of benefits. First, input from throughout the organisation improves the quality of the plans - good suggestions can come from any level in the organisation. Involvement in the planning process also enhances the overall understanding of the organisation's direction. Knowing the big picture can minimise friction between departments, sections, and individuals. For example, through planning, the sales department can understand and appreciate the objectives of the production department and their relationship to organisational objectives. Involvement in the planning process fosters a greater personal commitment to the plan; the plan becomes "our" plan rather than "their" plan. Positive attitudes created by involvement also improve overall organisational morale and loyalty.

Planning can also have positive effects on managerial performance. Studies have demonstrated that employees who stress planning earn very high performance ratings from supervisors. They have also shown that planning has a positive impact on the quality of work produced. While some have proved inconclusive, several studies have reported a positive relationship between planning and certain measures of organisational success such as profits and goals. One explanation that would fit all the findings to date is that good planning - as opposed to the mere presence or absence of a plan - is related to organisational success.

A final reason for planning is the mental exercise required to develop a plan. Many people believe the experience and knowledge gained throughout the development of a plan requires managers to think in a future- and contingency- oriented manner; this can result in great advantages over managers who are static in their thinking.

Just as there are reasons for engaging in the planning process, there are reasons why managers do not plan. Many managers become frustrated with the uncertainties associated with planning. Some take the attitude, "how can I predict the future" or "if I can't be exactly correct I'm not going to do it." Others view planning as something to do when there is nothing else to do. These managers

usually do not perceive any link between planning and performance. Yet another reason for avoiding planning is that it is hard work that usually requires considerable time and energy.

Formal Planning

All managers plan. The difference lies in the methods they employ and the extent to which they plan. Most planning is carried out on an informal or casual basis. This occurs when planners do not record their thoughts but rather carry them around in their heads. A formal plan is a written, documented plan developed through an identifiable process. The need for formal planning is not limited to large organisations; small ones can realise the same benefits. Formal planning processes can vary greatly from large to small organisations and even among similar-sized organisations. One might have a 2-page formal plan; another 100-page document. The appropriate degree of sophistication depends on the needs of the individual managers and the organisation itself. The environment, size, and type of business are factors that typically affect the planning needs of an organisation.

Who Should Plan?

Formal planning is not reserved strictly for top level managers. Formal planning should be practiced by all levels of management - from first-line supervisors right up to the chairman of the board. The detail and type of plan varies greatly with the level and responsibility of the manager. For example, first-line supervisors normally prepare plans to aid in meeting the objectives of their departments. Such plans might include a day-by-day breakdown of what needs to be done. Middle-level managers also prepare plans to aid in the accomplishment of their objectives. Normally, the plans of middle management are for a longer duration and are more comprehensive than those of managers working beneath them. Top level managers develop plans that guide the entire organisation.

Functional Plans

Plans are often classified by function or use. The most frequently encountered types of functional plan are sales and marketing plans, production plans, financial plans, and personnel plans. Sales and marketing plans are for developing new products/services and selling both present and future products/services. Production plans deal with producing the desired products/services on a timely schedule. Financial plans primarily deal with meeting the financial commitments and capital expenditures of the organisation. Personnel plans relate to the human resource needs of the organisations. Many functional plans are inter-related and inter-dependent. For example, a financial plan would be dependent on production, sales, and personnel plans.

Planning Horizon: Short Range, Intermediate, and Long Range

The length of the planning horizon varies somewhat from industry to industry, depending on the specific environment and activity. What may be long range when operating in a rapidly changing environment, such as the electronics industry, may be short range when operating in a relatively static environment such as the brick-manufacturing industry. In practice, however, short-range plans generally cover up to one year. Most long-range plans span at least three to five years, with some extending as far as 20 years into the future. While long- range planning is possible at any level in the organisation, it is primarily carried out at the top levels of the organisation. Intermediate plans cover that time span between short-range and long-range plans. From a pragmatic standpoint, intermediate plans generally cover from one to three or one to five years, depending on the horizon covered by the long-range plan. Usually intermediate plans are derived

from long-range plans, and short-range plans are derived from intermediate plans. For example, if the long-range plan calls for a 40 percent growth in sales by the end of five years, the intermediate plan should outline the necessary steps to be taken over the time span covering one to five years. Short-range plans would outline what actions are necessary within the next year.

It should be noted that some managers refer only to long-range and short- range plans, omitting intermediate plans. In this context, long range usually refers to plans starting at the end of the current year and extending into the future.

Operational versus Strategic Plans

Strategic planning is analogous to top level, long-range planning. It is the planning process applied at the highest levels of the organisation, covering a relatively long period and affecting many parts of the organisation.

Operations or tactical planning is short-range planning and concentrates on the formulation of functional plans. Production schedules and day-to-day plans are examples of operational plans. However, the distinctions between strategic and operations planning are relative, not absolute. The major difference is the level at which the planning is done: Strategic planning is primarily done by top level managers whereas operational planning is done by managers at all levels in the organisation and especially by middle- and lower-level managers.

Objectives

If you don't know where you're going, how will you know when you get there? Objectives are more specific and shorter range than are goals. Objectives are statements outlining what the organisation is trying to achieve; they give an organisation and its members direction and purpose. Few managers question the importance of objectives, only what the objectives should be.

Management is a form of work that involves co-ordinating an organisation's resources -land, labour, and capital - toward accomplishing organisational goals or objectives; the process centres around organisational objectives. Management cannot be properly practiced without pursuing specific objectives. Managers today and in the future must concentrate on where they and their organisations are headed.

It is also important to realise that managers and employees at all levels in an organisation should have objectives; everyone should know what he or she is trying to achieve. One key for organisational success is for the objectives at all different levels to mesh together.

A Cascade Approach

The key to ensuring that objectives properly relate to one another is the manner in which they are set. One desirable approach to setting objectives is to have the objectives "cascade" down through the organisation.

1. The objective-setting process begins at the top with a clear, concise statement of the central purpose or mission of the organisation.
2. Long-range organisational goals are formulated from this statement.
3. The long-range goals lead to the establishment of more, short-range performance objectives for the organisation. When tied to a specific time period, such as a year, these performance

objectives become the basis for, and an integral part of, the objectives of the chief executive and the top management team.

4. Derivative objectives are then developed for each major division or department.
5. Objectives are then established for the various sub-units in each major division or department.
6. The process continues on down through the organisational hierarchy.

This cascade approach to objective setting does not imply autocratic or "top- down" management. It merely ensures the objectives of individual units within the organisation are in phase with the major objectives of the organisation. It co-ordinates the entire objective-setting process.

Organisational Mission

An organisation's mission is actually the broadest and highest level of objectives. Mission defines the basic **purpose** or purposes of the organisation (for this reason, the terms mission and purpose are often used interchangeably). Basically, an organisation's mission outlines why the organisation exists. A mission statement usually includes a description of the organisation's basic products and/or services and a definition of its markets and/or sources of revenue.

Defining mission is crucial. It is also more difficult than one might imagine. Peter Drucker emphasises that an organisation's purpose should be examined and defined not only at its inception or during difficult times but also during successful periods. Drucker argues that an organisation's purpose is not determined by the organisation itself but by its customers. Customer satisfaction with an organisation's product or service defines the purpose more than does the organisation name, statutes, or articles of incorporation. Drucker outlines three questions that need to be answered to define an organisation's present business.

- First, management must identify the customers - where they are, how they buy, and how they can be reached.
- Second, management must know what the customer buys.
- Finally, what is the customer looking for in the product?

Management must also identify what the future business will be and what it should be. Drucker presents four areas to investigate.

- The first is market potential: What does the long-term trend look like?
- Second, what changes in market structure might occur because of economic developments, changes in styles or fashions, or competition? For example, how have oil prices affected the automobile market structure?
- Third, what possible changes will alter the customers' buying habits? What new ideas or products might create new customer demands or change old demands? Consider, for example, the impact of the mini-calculator on the sale of slide rules and hand-cranked adding machines.
- Finally, what customer needs are not being adequately served by available products and services?

Long-Range and Short-Range Objectives

Long-range objectives generally go beyond the current fiscal year of the organisation. Long-range objectives must support and not conflict with the organisational mission. But they may be quite different from the organisation mission and still support it. For instance, the organisational mission

of a fast- food restaurant might be to provide rapid, hot food service to a certain area of the city. One long-range objective might be to increase sales to a specific level within the next four years. This objective is quite different from the organisational mission; but it still supports the mission.

Short-range objectives should be derived from an in-depth evaluation of long-range objectives. Such an evaluation should result in a listing of priorities of the long-range objectives. Then short-range objectives an be set to help achieve the long-range objectives.

All levels within the organisation should set objectives based on the long- range and short-range objectives of the organisation. Objectives at any level must be co-ordinated with, and subordinated to, the objective of the next higher level. All objectives are then synchronised and not working against each other.

Short-range objectives usually cover up to one year, intermediate objectives usually cover one to three or one to five years into the future, and long-range objectives cover over three or over five years into the future.

Objectives should be clear, concise, and quantified when possible. Affected personnel should clearly understand what is expected. Normally, multiple objectives should be used to reflect the desired performance of a given organisational unit or person. From a top-level perspective, objectives should span all major areas of the organisation. A problem with one overriding objective is that it is often achieved at the expense of other desirable objectives. For example, if production is the only objective, quality might suffer in order to realise maximum production. While objectives in different areas may serve as checks on each other, they should be reasonably consistent with each other.

Objectives should be dynamic; they should be re-evaluated as the environment and opportunities change. Objectives for organisations normally fall into one of four general categories: (1) profit oriented, (2) service to customers, (3) employee needs and well-being, and (4) social responsibility. Even non-profit organisations must be concerned with profit in the sense that they generally must operate within a budget. Another scheme for classifying organisational objectives is (1) primary, (2) secondary, (3) individual, and (4) societal. Primary objectives are those that relate directly to profit. Secondary objectives are those that relate to specific units of the organisations (i.e., departmental objectives). Individual objectives relate directly to the organisation's employee. The final category, societal, refers to those objectives that relate to the local, national and global communities. Listed below are potential areas for establishing objectives in most organisations:

1. Profitability: the degree to which the firm is attaining an acceptable level of profits; usually expressed in terms of profits before or after taxes, return on investment, earnings per share, or profit-to-sales ratios.
2. Markets: reflects the firm's position in its marketplace, expressed in terms of share of the market, unit volume in sales, or niche in the industry.
3. Productivity: measures the efficiency of internal operations expressed as a ratio of inputs to outputs, such as number of items or services produced per unit of time.
4. Product: describes the introduction or elimination of products or services; expressed in terms of when a product or service will be introduced or dropped.

5. Financial resources: reflects goals relating to the funding needs of the firm; expressed in terms of capital structure, new issues of common stock, cash flow, working capital, dividend payment, and collection periods.
6. Physical facilities: describes the physical facilities of the firm; expressed in terms of square metres of office or plant space, fixed costs, units of production, or other similar measurements.
7. Research and innovation: reflects the research, development, and/or innovation aspirations of the firm; usually expressed in terms of monetary terms to be expended.
8. Organisation structure: describes objectives relating to changes in the organisational structure and related activities; expressed in terms of a desired future structure or network of relationships.
9. Human resources: describes the human resource, assets of the organisations; expressed in terms of absenteeism, tardiness, number of grievances, and training.
10. Social responsibility: refers to the commitments of the firm regarding society and the environment; expressed in terms of types of activities, number of days of service, or financial contributions.

Management by Objectives (MBO)

One approach to setting objectives that has enjoyed considerable popularity is the concept of management by objectives. MBO is a philosophy based on converting organisational objectives into personal objectives. It assumes that establishing personal objectives elicits employee commitment, which leads to improved performance. The MBO process is has also been called management by results, goals and control, work planning and review, and goals management. All these programs are similar and follow the same basic process.

The objective-setting process in MBO is best accomplished by using the cascade approach to objective setting. Setting objectives from top to bottom creates an integrated hierarchy of objectives throughout the entire organisation. It ensures that the various levels within the organisation have a common direction.

In MBO, the objective-setting process requires involvement and collaboration among the various levels of the organisation; this joint effort has beneficial results. First, people at each level become more aware of organisational objectives. The better they understand the organisation's objectives, the better they see their roles in the total organisation. Second, the objectives for an individual are jointly set by the person and the superior; there are give-and-take negotiating sessions between them: Achieving self-formulated objectives can improve motivation and, thus, job performance. It should be stressed that the subordinate should genuinely be involved in setting his or her objectives as opposed to simply agreeing with objectives the boss has set.

Setting objectives in MBO is not always easy, and problems occur frequently. Often the most difficult problem is deciding the specific areas in which to set objectives. Objectives should be set in areas that are vital to the success of the organisational unit being measured. These areas are referred to as key result areas. In general, every key result area should be covered by an objective. Key result areas exist for the organisation as a whole and for the different subunits of the organisation. For example, a corporate sales objective would represent a key result area for the organisation as a whole. On the other hand, the number of boxes packed per day would represent a

key result area for the packing department. A practical way to identify key result areas is to answer the question:

What areas should be monitored to determine whether the organisation (or the organisational unit) is doing a good, average, or poor job? In other words, key result areas should reflect the performance of the organisation or the organisational unit in question.

A rule of thumb is for each organisational unit to have somewhere between three and seven key result areas. Experience has shown that more than one or two key result areas are usually necessary to capture the essence of an organisational unit's performance. At the same time, however, more than seven key result areas often become unwieldy.

After the objectives have been jointly established, a plan of action for achieving them should be developed. This involves the following:

1. Determine the major activities necessary to accomplish the objectives.
2. Establish sub-activities necessary for accomplishing the major activities.
3. Assign primary responsibility for each activity and sub-activity.
4. Estimate time requirements necessary to complete each activity and sub-activity.
5. Identify additional resources required for each activity and sub-activity.

After establishing objectives and outlining the actions necessary to accomplish them, individuals are allowed to pursue their objectives essentially in their own manner. Therefore, MBO is largely a system of self-control. However, there are policy constraints on individuals; but basically, people achieve goals through their own abilities and effort. The Table below gives guidelines for establishing individual objectives.

Guidelines for Establishing Individual Objectives

1.	Adapt your objectives directly to organisational goals and strategic plans. Do not assume they support higher-level management objectives.
2.	Quantify and target the results whenever possible. Do not formulate objectives whose attainment cannot be measured or at least verified.
3.	Test your objectives for challenge and achievability. Do not build in cushions to hedge against accountability for results.
4.	Adjust the objectives to the availability of resources and the realities of organisational life. Do not keep your head either in the clouds or in the sand.
5.	Establish performance reports and milestones that measure progress toward the objective. Do not rely on instinct or crude benchmarks to appraise performance.
6.	Put your objectives in writing and express them in clear, concise, and unambiguous statements. Do not allow them to remain in loose or vague terms.
7.	Limit the number of statements of objectives to the most relevant key result areas of your job. Do not obscure priorities by stating too many objectives.
8.	Communicate your objectives to your subordinates so that they can formulate their own job objectives. Do not demand they do your goal setting for you.
9.	Review your statements with others to assure consistency and mutual support. Do not fall into the trap of setting your objectives in a vacuum.
10.	Modify your statements to meet changing conditions and priorities.
11.	Do not continue to pursue objectives that have become obsolete.

Periodic progress reviews are an essential ingredient of MBO. This includes giving each employee feedback on actual performance as compared to planned performance (objectives). The importance of this personal feedback cannot be overestimated.

The manner in which the feedback is given is important. If it is hostile, performance may be reduced. Reviews should not be used to degrade the individual. To summarise, an MBO system in its simplest form must meet the following three minimum requirements:

1. Individual objectives are jointly set by the subordinate and the superior.
2. Individuals are periodically evaluated and receive feedback concerning their performance.
3. Individuals are evaluated and rewarded on the basis of objective attainment.

Many organisations proudly proclaim they have successfully implemented MBO when, in fact, they have met very few of the actual requirements. Research has shown that the success rate of MBO programs is dismal. However, a close investigation of the failures often reveals the fault is not in the program itself but rather the manner in which it was implemented. For example, a manager unilaterally sets objectives for his or her subordinates and then asks them if they agree with the stated objectives. Feeling pressure to agree, subordinates usually go along with the boss. Under these conditions, the spirit of MBO has been violated and the objectives have not been jointly set. Another violation occurs when subordinates' performance is reviewed very infrequently. True MBO requires regular and periodic evaluations and feedback. A third frequently encountered misuse of MBO is when rewards are not related to objective attainment. An organisation that gives across-the-board pay raises to all of its employees is an example of this. Studies have also shown that a major reason for the failure of an organisation wide MBO system is the absence of a fully committed and involved top management). If top management does not personally get involved employees usually do not view the system as important and it usually fails. An ideal MBO system permeates the entire organisation, starting at the top of the organisation and cascading downward to its lowest levels. However, it is possible for a manager to use MBO is his or her department even when the system is not being used throughout the organisation

Policies

Policies are broad, general guides to action that constrain or direct objective attainment. Policies do not tell organisational members exactly what to do, but they do establish the boundaries within which they must operate. For example, a policy of answering all written customer complaints in writing within 10 days" does not tell a manager exactly how to respond, but it does say it must be done in writing within 10 days. Policies create an understanding among members or group that makes the actions of each member more predictable to other members.

Policy statements often contain the words 'to ensure", 'to follow', 'to maintain', 'to promote', 'to be', 'to accept', and similar verbs. For example, a Company XYZ, may have a policy "to accept all returns that are accompanied by a sales slip." Such a policy is a general guideline for pursuing company objectives related to profit and sales.

Policies exist at all levels of an organisation: A typical organisation has some policies that relate to everybody in the organisation and some that relate only to certain parts of the organisation. A policy such as "this company will always try to fill vacancies at all levels by promoting from within" would relate to everyone in the organisation. On the other hand, the previously described

policy requiring that all customer complaints be answered in writing relates only to personnel in a specific department.

Procedures and Rules
Procedures and rules differ from policies only in degree. In fact, they may be thought of as low-level policies. A **procedure** is a series of related steps or tasks expressed in chronological order for a specific purpose. Procedures define in step-by-step fashion the methods through which policies are achieved. Procedures emphasise details. They outline precisely how a recurring activity must be accomplished. Procedures allow for little flexibility and deviation. A company's policy may be to accept all customer returns submitted within one month of purchase; company procedures would outline exactly how a return should be processed. Well-established, formal procedures are often known as standard operating procedures (SOPs).

Rules require specific and definite actions to be taken or not to be taken in a given situation. Rules leave little doubt about what is to be done. They permit no flexibility and deviation. Unlike procedures, rules do not have to specify sequence. For example, "No smoking in the conference room" is a rule.

Procedures and rules are actually subsets of policies. All provide some kind of guidance. The differences lie in the range of applicability and the degree of flexibility. A no-smoking rule is much less flexible than a procedure for handing customer complaints; the latter is likewise less flexible than a hiring policy.

Integrating Objectives, Policies, and Plans
Organisational objectives, policies, and plans are not mutually exclusive components of the management process. They are highly interdependent and inseparable. One cannot effectively pursue objectives without first knowing what they are and what policy guidelines must be followed. The importance of clear and sound objectives cannot be overstated. As the old saying goes, "If you don't know where you're going, any road will get you there."

TOOLS FOR PLANNING AND DECISION MAKING
To plan managers must make assumptions about future events. Planners must develop forecasts of probable future circumstances. Forecasting is the process of developing assumptions or premises about the future that managers can use in planning or decision making.

Sales and Revenue Forecasting
- Sales Forecasting – The prediction of future sales
- Revenue Forecasting – The prediction of future revenues from all sources

Technological Forecasting – The prediction of what future technologies are likely to emerge and when they are likely to be economically feasible.

Other Types of Forecasting: Forecasting Techniques
Time-Series Analysis A forecasting technique that extends pasts information into the future through the calculation of a best-fit line.

Causal Modelling A group of different techniques that determine causal relationships between different variables

***Regression models*:** Used to predict one variable (called the dependent variable) on the basis of known or assumed other variables (called in dependent variables). For example, we might predict future sales based on the values of price, advertising, and economic levels.

***Econometric models*:** Make use of several multiple-regression equations to consider the impact of major economic shifts.

***Economic indicators*:** Various population statistics, indexes, or parameters that predict organisationally relevant variables such as discretionary income. Examples include cost-of-living index, inflation rate, and level of unemployment.

MANAGERIAL DECISION MAKING

Decision making versus Problem Solving

The term decision making and problem solving are often confused and therefore, need to be clarified. **Decision making**, in its narrowest sense, is the process of choosing from among various alternatives. A problem is any deviation from some standard or desired level of performance. **Problem solving**, is then the process of determining the appropriate responses or actions necessary to alleviate a problem. Problem solving necessarily involves decision making since all problems can be attacked in numerous ways and the problem solver must decide which is best. On the other hand, all decisions do not involve problems (such as a person sorting fruit or vegetables; depositing your money in a bank savings account). However, from a practical perspective, almost all managerial decisions do involve solving or at least avoiding problems. Also from a practical perspective it is usually not necessary to distinguish between managerial decision making and managerial problem solving.

Characteristics of Managerial Decisions

Managers face problems constantly. Some problems that require a decision are relatively simple; others seem overwhelming. Some demand immediate action, while others take months or even years to unfold.

Actually, managers often ignore problems. For several reasons, they avoid taking action. First, managers can't be sure how much time, energy, or trouble lies ahead once they start working on a problem. Second, getting involved is risky; tackling a problem but failing to solve it successfully can hurt the manager's track record. Third, because problems can be perplexing, it is easier to procrastinate or get busy with less demanding activities.

It is important to understand why decision making can be so challenging. Figure 1 illustrates several characteristics of managerial decisions that contribute to their difficulty and pressure. Most managerial decisions lack structure and entail risk, uncertainty, and conflict.

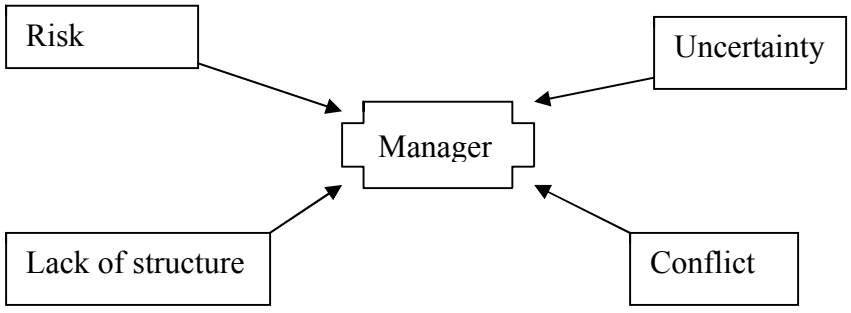


Figure 1 Characteristics of managerial decisions

Lack of structure

Lack of structure is the usual state of affairs in managerial decision making. Although some decisions are routine and clear-cut, for most there is no automatic procedure to follow. Problems are novel and unstructured, leaving the decision maker uncertain about how to proceed.

A well-known distinction illustrating this point is between programmed and non-programmed decisions. **Programmed decisions** have been encountered and made before. They have objectively

correct answers and can be solved by using simple rules, policies, or numerical computations. If you face a programmed decision, there exists a clear procedure or structure for arriving at the right decision. For example, if you are a small business owner and must decide the amounts for your employees' pay cheques, you can use a calculator and if the amounts are wrong, your employees will prove it to you.

If most important decisions were programmed, managerial life would be much easier. But managers typically face non-programmed decisions: new, novel, complex decisions having no certain outcomes. There are a variety of possible solutions, all of which have merits and drawbacks. The decision maker must create or impose a method for making the decision; there is no predetermined structure on which to rely.

Uncertainty and risk

If you have all the information you need, and can predict precisely the consequences of your actions, you are operating under a condition of **certainty**. Managers are expressing their preference for certainty when they are not satisfied hearing about what *may* have happened or *might* happen and insist on hearing what *did* or *will* happen. But perfect certainty is rare. For important managerial decisions, uncertainty is the rule.

Uncertainty means the manager has insufficient information to know the consequences of different actions. Decision makers may have strong opinions – they may feel sure of themselves – but are still operating under conditions of uncertainty if they lack pertinent information and cannot estimate the likelihood of different results of their actions.

When you estimate the likelihood of various consequences, but still do not know with certainty what will happen, you are facing **risk**. Risk exists when the probability of an action being successful is less than 100 percent. If the decision is the wrong one, you may lose money, time, reputation, or other important assets.

Risk, like uncertainty, is a fact of life in managerial decision making. But this is not the same as taking a risk. Whereas it sometimes seems as though risk takers are admired, and that entrepreneurs and investors thrive on taking risks, the reality is that good decision makers prefer to *avoid* or *manage* risk. This means that while they accept the fact that consequential decisions entail risk, they do everything they can to anticipate the risk, minimise it, and control it.

Conflict

Important decisions are even more difficult because of the conflict managers face. **Conflict**, which exists when the manager must consider opposing pressures from different sources, exists at two levels.

First, individual decision makers experience psychological conflict when several options are attractive, or even none of the options is attractive. For instance, a manager may have to decide whom to lay off, when he doesn't want to lay off anyone. Or he may have three promising job applicants for one position – but choosing one means he has to reject the other two.

Second, conflict arises between individuals or groups. The chief financial officer argues in favour of increasing long-term debt to finance an acquisition. The chief executive officer, however,

prefers to minimise such debt and find the funds elsewhere. The marketing department wants more product lines to sell to its customers, and engineers want higher quality products. But production people want to lower costs by having longer production runs of fewer products with no changes. Management wants to enforce some rigid work rules, while labour seeks looser rule enforcement. Few decisions are unanimous or uncompromised.

The Stages of Decision Making

The decision making process moves through six stages:

1. Identifying and diagnosing the problem
2. Generating alternative solutions
3. Evaluating alternatives
4. Making the choice
5. Implementing the decision
6. Evaluating the decision

Identifying and diagnosing the problem

The first stage in the decision making process is to recognise that a problem exists and must be solved. Typically, a manager realises some discrepancy between the current state (the way things are) and a desired state (the way things ought to be). Such discrepancies – say in organisational or departmental performance – may be detected by comparing (1) current performance against past performance, (2) the current performance of other organisations or departments, or (3) future performance as determined by plans and forecasts.

Recognising that a problem exists is only the beginning of this stage. The decision maker also must want to do something about it and must believe that the resources and abilities necessary for solving the problem exist. Then the decision maker must dig in deeper and attempt to diagnose the true cause of the problem symptoms that surfaced.

For example, a sales manager knows that sales have dropped drastically. If he is leaving the company soon or believes the decreased sales volume is due to the economy (which he can't do anything about), he won't take further action. But if he does try to solve the problem, he should not automatically reprimand his sales staff, add new people, or increase the advertising budget. He must *analyse* why sales are down and then develop a solution appropriate to his analysis. Asking why, of yourself and others, is essential to understanding the real problem.

Useful questions to ask and answer in this stage include:

- Is there a difference between what is typically happening and what should be happening?
- How can you describe the deviation, as specifically as possible
- What is/are the cause(s) of the deviation?
- What specific goals should be met?
- Which of these goals are absolutely critical to the success of the decision?

Generating alternative solutions

In the second stage, problem diagnosis is linked to the development of alternative courses of action aimed at solving the problem. Managers generate at least some alternative solutions based on past experiences.

Solutions range from ready made to custom made. Decision makers who search for **ready-made solutions** use ideas they have seen or tried before or follow the advice of others who have faced similar problems. **Custom made solutions**, on the other hand, must be designed for specific problems. This technique requires combining ideas into new, creative solutions. For example, the Sony Walkman was created by combining two existing products: earphones and a tape player. Potentially custom made solutions can be devised for any challenge.

Choosing a ready made alternative is much easier than designing a custom made solution. Therefore, most decision makers make use of the ready made approach, sometimes even when the ready made alternative is inappropriate. If this approach fails to deliver acceptable solutions or results, the harder work of devising a unique solution must begin. For important, irreversible decisions, custom made alternatives should be developed because they are more likely to lead to higher quality solutions.

Useful questions at this stage include:

- Is there a particular alternative to be evaluated?
- Are there others we should consider?
- Who can help us by providing ideas?
- How can we generate additional alternatives creatively?

Evaluating alternatives

The third stage involves determining the value or adequacy of the alternatives that were generated. Which solution will be the best?

Too often, alternatives are evaluated with little thought or logic. Alternatives should be evaluated carefully. It is important to predict the consequences that will occur if the various options are put into effect.

Managers should consider several types of consequences. They must attempt to predict the effects on financial or other performance measures. But there are other, less clear cut consequences to address. Decisions set are precedent; will this precedent be a help or a hindrance in the future? Also, the success or failure of the decision will go into the track records of those involved in making it.

Refer again to your original goals, defined in the first stage. Which goals does each alternative meet, and fail to meet? Which alternatives are most acceptable to you and to other important stakeholders? If several alternatives may solve the problem, which can be implemented at the lowest cost? If no alternative achieves all your goals perhaps you can combine two or more of the best ones.

Key questions here are:

- Is our information about our alternatives complete and current? If not, can we get more and better information?
- Does the alternative meet our primary objectives?
- What problems could we have if we implement the alternative?

Results cannot be forecast with perfect accuracy. But sometimes decision makers can build in safeguards against an uncertain future by considering the potential consequences of several different scenarios. Then they generate contingency plans: alternative courses of action that can be implemented based on how the future unfolds.

Making the choice

Once you have considered the possible consequences of your options, it is time to make your decision. Important concepts here are maximising, satisficing, and optimising.

To **maximise** is to make the best possible decision. The maximising decision realises the greatest positive consequences and the fewest negative consequences. In other words, maximising results in the greatest benefit at the lowest cost, with the largest expected total return. Maximising requires searching thoroughly for a complete range of alternatives, carefully assessing each alternative, comparing one to another, and then choosing or creating the very best.

To **satisfice** is to choose the first option that is minimally acceptable or adequate; the choice appears to meet a targeted goal or criterion. When you satisfice, you compare your choice against your goal, not against other options. Satisficing means a search for alternatives stops at the first one that is okay. Commonly, people do not expend the time or energy to gather complete information. Instead, they make the expedient decision based on readily available information. Satisficing is sometimes a result of laziness; other times, there is no other option because time is short, information is unavailable, or other constraints make it impossible to maximise.

Let's say you are purchasing new equipment and your goal is to avoid spending too much money. You would be maximising if you checked out all your options and their prices, and then bought the cheapest one that met your performance requirements. But you would be satisficing if you bought the first one you found that was within your budget and failed to look for less expensive options.
Optimising means that you achieve the best possible balance among several goals. Perhaps, in purchasing equipment, you are interested in quality and durability as well as price. So, instead of buying the cheapest piece of equipment that works, you buy the one with the best combination of attributes, even though there may be options that are better on the price criterion and others that are better on the quality and durability criteria. The same idea applies to achieving business goals: one marketing strategy could maximise sales, while a different strategy might maximise profit. An optimising strategy is the one that achieves the best balance among multiple goals.

Implementing the decision

The decision-making process does not end once a choice is made. The chosen alternative must be implemented. Sometimes the people involved in making the choice must put it into effect. At other times, they delegate the responsibility for implementation to others, such as when a top management team changes a policy or operating procedures and has operational managers carry out the change.

Those who implement the decision must understand the choice and why it was made. They also must be committed to its successful implementation. These needs can be met by involving those people in the early stages of the decision process.

Managers should plan implementation carefully. Adequate planning requires several steps:

1. Determine how things will look when the decision is fully operational.
2. Chronologically order, perhaps with a flow diagram, the steps necessary to a fully operational decision.
3. List the resources and activities required to implement each step.
4. Estimate the time needed for each step.
5. Assign responsibility for each step to specific individuals.

Decision makers should assume that things will not go smoothly during implementation. It is very useful to take a little extra time to identify potential problems and identify potential opportunities. Then, you can take actions to prevent problems and also be ready to seize on unexpected opportunities. Useful questions are

- What problems could this action cause?
- What can we do to prevent the problems?
- What unintended benefits or opportunities could arise?

Always expect the unexpected. It is better not to learn about unexpected issues in the implementation stage but to do a thorough job thinking through the earlier stages in the decision-making process.

Evaluating the decision

The final stage in the decision-making process is evaluating the decision. This means collecting information on how well the decision is working. Quantifiable goals—a 20 percent increase in sales, a 95 percent reduction in accidents, 100 percent on-time deliveries—can be set before the solution to the problem is implemented. Then objective data can be gathered to accurately determine the success (or failure) of the decision.

Decision evaluation is useful whether the feedback is positive or negative. Feedback that suggests the decision is working implies that the decision should be continued and perhaps applied elsewhere in the organisation. Negative feedback, indicting failure means that either

1. implementation will require more time, resources, effort, or thought or
2. the decision was a bad one

If the decision appears inappropriate, it's back to the drawing board. Then the process cycles back to the first stage: (re)definition of the problem. The decision making process begins anew, preferably with more information, new suggestions, and approach that attempts to eliminate the mistakes made the first time around.

The best decision

How can managers tell whether they have made the best decision? One approach is to wait until the results are in. But if the decision has been made but not yet implemented? While nothing can guarantee a "best" decision, managers should at least be confident that they followed proper procedures that will yield the best decision under the circumstances. This means that the decision makers were appropriately vigilant in making the decision. **Vigilance** occurs when the decision makers carefully and conscientiously execute all six stages of decision making, including making provisions for the implementation and evaluation.

Types of Decisions

Managers and other employees make many different types of decisions. Three types of decisions often used include: strategic, operational, and problem-solving decisions.

Strategic decisions are those decisions that have a major impact on the general direction of the firm. These decisions are often carefully considered and may involve millions of dollars of investment. They may change the way a business competes. They may involve the introduction of new products or the acquisition of another company. They may require hiring hundreds of employees or laying off thousands. Perhaps the most extreme of all is the decision to sell or close the business. The characteristics of strategic decisions are as follows:

- Have long-range impact.
- Require careful analysis of the firm and its environment.
- Are often the result of a strategic planning process.
- May involve millions of dollars.
- Are designed to capture opportunities or offset competitive weakness
- Involve top management, including board of directors

The most significant characteristic of strategic decisions are long range decisions made by top management. Others in the organisation will have input, but the magnitude of the decision dictates that managers at the very highest level be involved. Strategic decisions involve s structured, analytical process that carefully considers as much information as can possibly be gained. These decisions are typically made slowly, after careful study.

Operational decisions are those that affect the day-to-day actions of the business. Characteristics of operational decisions are:

- May or may not involve large amounts of capital
- Deal with specific situations within an organisation
- Are made by experienced managers in the area
- Often have standard operating procedures available to guide decisions
- Require less analysis than strategic decisions do

Operational decisions may or may not involve large amounts of capital, but they tend to deal with specific situations and are made by managers who have considerable experience in the area affected by the decision.

Operational decisions might include the volume of products to be produced this month, the selection of a new supplier to replace one whose quality is unacceptable, the selection of new employees to replace those who have left the company or who have been promoted, the amount of' money budgeted for maintenance of equipment, or the selection of TV stations on which to advertise.

Another characteristic is that operational decisions are frequently made within the boundaries of established policies and procedures. There are often standard operating procedures (SOPs) in place to guide the decision making. This structure eases the decision-making process. It means: that managers do not need to do a significant study of the situation each time a new decision must be

made. Only those decisions that fall outside the purview of the SOP need substantial amounts of analysis.

Problem Solving Decisions

Problem solving is decision making aimed at correcting an adverse situation that has developed. The orientation of problem solving is how to fix something that is: wrong. Examples abound. A key staff employee decides to leave the company abruptly, and the company is left with no one who knows how to do billing correctly. Two employees who previously were cooperative suddenly argue constantly. A firm's accountant presents a department manager with information suggesting that the department's telephone expenses were over budget by 25 percent.

The common element of each of these examples is that the situation is some adverse deviation from the norm. Typical characteristics of problem-solving decisions include:

- Situation is a deviation from normal operations
- May be minor or of crisis proportions
- Often need quick decisions to find solutions acceptable to all involved
- There may be no easy solution

Problems to be solved are sometimes of crisis proportions and sometimes minor. The key to problem solving is to reach a decision that leaves the organisation in the best possible position, given the situation at hand. The task - is to analyse the situation as carefully, but quickly, as possible and find a solution to the problem that is acceptable to all involved.

In some cases, there are no good solutions. Most experienced managers know that occasionally a situation arises that simply cannot be completely resolved. Here the manager must carefully decide which of a set of poor alternatives comes closest to solving the problem, knowing that not everyone will be happy. This is unfortunate, but it is the reality of the workplace. For example, suppose your company has a hiring freeze; top management has dictated that absolutely no new employees will be hired. One of your long-time workers decides to take early retirement. This will create a void in your department, and work will have to be spread among the remaining workers. But the reality is that you will not get a replacement and you must simply live with the problem. You are forced to reallocate work among existing employees to take up the slack.

Sometimes the best decision is to leave the problem unresolved for now in hopes that conditions may change. For example, suppose you work in a job that requires you to spend hours each day working on a computer. Your computer seems to be increasingly slow and locks up more often than it should. You report this problem to your supervisor, indicating your frustration with the computer. The supervisor could turn in a repair request immediately to have your computer fixed, but your entire department is scheduled to get new computers in two months. Thus, it does not make sense to have your computer repaired. You are told to keep using it, back up files frequently, and hope it doesn't crash completely before the new computers arrive.

Participation in Decision Making

Most managers have opportunities to involve their subordinates and others in the decision-making process. One pertinent question is, Do groups make better decisions than individuals? Another is, When should subordinates be involved in making managerial decisions?

Group Decision Making

Everyone knows the old axiom that two heads are better than one. Empirical evidence generally supports this view—with a few minor qualifications. Group performance is frequently better than that of the average group. Similarly, groups can often be successfully used to develop innovative and creative solutions to problems. Groups also take longer to solve problems than does the average person." Thus, group decisions are generally better when avoiding mistakes is more important than speed.

Group performance is generally superior to that of the average group member for two basic reasons. First, the sum total of the group's knowledge is greater; second, the group has a much wider range of alternatives in the decision process. Group decision making also has other benefits. It facilitates and increases acceptance of the decision by group members, especially when a change is being implemented. A more complete understanding of both the decision and alternative solutions results from group decision making. This is especially helpful when those who must implement the decision participate in the process.

However, some potential drawbacks can greatly limit the effectiveness of group decision making. One person may dominate or control the group. This occurs frequently when the president or other higher-ups in the organisation take part in the decision process. Because of their presence, many other members become inhibited. The social pressures of conformity can also inhibit group members.

Competition can develop within the group to such an extent that winning an issue becomes more important than the issue itself. A final hazard results from the dynamics of group decision making. Groups tend to accept the first potentially positive solution and give little attention to others.

One other trait of group decision making compares the risk that people will take alone and in a group. Laboratory experiments have shown that unanimous group decisions are consistently more risky than the average of the individual decisions.' This is somewhat surprising since grot pressures often inhibit the members. Possibly people feel less responsible for the outcome of a group decision than when they act alone. More recent research has found that groups make decisions best described as more polar than do individuals acting alone.' "More polar" means that groups tend to make decisions that are more extreme than they would make as individuals. The Table below summarises the positive and negative aspects of group decision making.

Positive and Negative Aspects of Group Decision Making

Positive Aspects	Negative Aspects
The sum total of the groups knowledge is greater	One individual may dominate and/or control the group
The group possesses a much wider range of alternatives in the decision process	Social pressures to conform can inhibit group members.
Participation in the decision-making process increases the acceptance of the decision by group members	Competition can develop to such an extent that winning becomes more important than the issue itself.
Group members better understand the decision and the alternatives considered.	Groups have a tendency to accept the first potentially positive solution, while giving little attention to other possible solutions.

Selecting the Decision-Making Style

One key to effective decision making is the ability to select the appropriate decision-making style for each decision faced. Should group decision making be used whenever possible? Should employees be consulted in all decisions or only in certain situations? Research and practice show that no single style of decision making works best in all situations. Successful leaders learn to match the appropriate decision-making style with the situation. The Table below depicts the types of management decision styles.

Types of Management Decision Styles

Al:	The manager makes the decision alone, with little or no input from subordinates (the A stands for autocratic).
All:	The manager asks subordinates for information that he or she needs to make the decision, but still makes the decision alone. Subordinates may or may not be informed of the decision. The role played by subordinates is clearly one of providing information as opposed to generating or evaluating alternative solutions.
Cl:	The manager shares the situation with selected subordinates and asks them individually for information and advice. The manager still makes the final decision, which may or may not reflect the subordinates' influence. (The C stands for consultative).
CII:	The manager meets with subordinates as a group to discuss the situation. Information is freely shared, although the manager still makes the final decision, which may or may not reflect the subordinates' influence.
GIl:	The manager and subordinates meet as a group and freely share information. The entire group makes the decision (The G stands for group).

Creativity in Decision Making

Creative people tend to exhibit independent judgment, personal complexity, a preference for complexity over simplicity, a preference for ambiguity over clarity, self-assertiveness, and a reluctance to control impulses.

1. Divergent production is the ability to generate numerous alternatives for accomplishing an objective.
2. Transformation is the ability to reinterpret and reorganize information in order to produce a solution.

The creative process consists of a series of steps that tend to occur in sequence.

1. Saturation involves the conscious analysis of a problem.
2. Incubation involves both conscious and unconscious thought processes. Illumination refers to the conscious awareness of an essential element of the solution.
3. Elaboration involves expanding or refining what is known about the problem.
4. Creativity is actually the discovery of a unique solution.

Several techniques have been developed to enhance creative decision making.

1. ***Brainstorming*** is designed to produce numerous ideas in a short period of time through the use of a classroom environment in which ideas are encouraged and criticism is not allowed.
2. ***Synectics*** employs a group containing at least one technical expert to review a problem and suggest a creative solution.

3. The ***Delphi technique*** seeks to develop a range of possible alternatives that reflect the collective judgment of the participants.
4. ***Fish-bowling*** requires a decision-making group to interview a series of "experts" prior to making decision.
5. ***Didactic interaction*** requires group members to list the pros and cons of a go/no-go type of decision and reach a final decision by switching roles during discussion.

Why Do Managers Make Poor Decisions?

Poor decisions may generally be traced to one or more common errors.

1. Adequate time may not be allowed for consideration or analysis.
2. Goals may not be clearly defined.
3. Unreliable sources of information may be employed.
4. Managers may make "safe" decisions because they fear the consequences of bold action.
5. Symptoms may dominate discussions rather than causes.
6. Intuition may be allowed to outweigh evidence.
7. The decision, once made, may not be implemented.

Suboptimal decisions may result in spite of the best efforts of the manager.

1. Bounded rationality imposes limits on a decision within which the manager must work.
2. Sub-optimisation refers to the tendency to put the interests of the manager's specific department above those of the company as a whole.
3. Unforeseen environmental changes may cause less-than-optimal decisions.

Tools for Decision making

- Forecasting models
- Moving averages
- Decision Trees
- Decisions under uncertainty, risk, conflict etc

Synectics: A method of problem-solving, esp. by groups, which seeks to utilize creative thinking and imaginative correlation.
Delphi: A technique of using questionnaires to arrive at consensual judgements, decisions, etc.
Didactic: Having the character or manner of a teacher; intended to instruct; having instruction as an ulterior purpose.

ORGANISING, LEADING AND STAFFING

ORGANISING

What is organising?

Organising involves the arrangement of human and physical resources in a structure designed to co-ordinate and channel effort toward the accomplishment of the firm's strategic plan. The process of organising is the grouping of activities necessary to attain common objectives and the assignment of each grouping to a manager who has the authority necessary to supervise the people performing the activities. Thus organising is basically a process of ***division of labour*** accompanied by appropriate ***delegation*** of ***authority***. Proper organising results in the better use of resources.

Why Organise?

- Organising is done to establish lines of authority. Clear lines of authority create order within a group. Absence of authority almost always leads to chaotic situations, where everyone is telling everyone else what to do.

- Organising improves the efficiency and quality of work through synergism. Synergism occurs when individuals or groups work together to produce a whole greater than the sum of the parts. For example, synergism results when three people working together produce more than three people working separately. Synergism can result from division of labour or from increased co-ordination, both of which are products of good organisation.

- Organising improves communication. A good organisation structure clearly defines channels of communication among the members of the organisation. Such a system also ensures more communication.

Organising Concepts

The two key concepts of organising are design and structure. Design in this context, implies that managers make a conscious effort to predetermine the way employees will do their jobs.

Organisational structure refers to relatively stable relationships and aspects of the organisation. Organisational structure may be thought of as the anatomy of the organisation, providing a foundation within which the organisation functions. The structure of an organisation, similar to the anatomy of a living organism, acts as a framework within which the dynamic activity and process of people doing work takes place.

The idea of structure as a framework "focuses on the differentiation of positions, formulation of rules and procedures, and prescriptions of authority. In this context, the purpose of structure is to regulate or at least reduce the uncertainty regarding the behaviour of individual employees. An organisation chart uses a series of boxes connected with one or more lines to graphically represent an organisation's structure.

Division of labour (or Specialisation)

Division of labour involves breaking down a task into its most basic elements, training workers performing specific duties, and sequencing activities so that one person's efforts build on another.

Labour can be divided either vertically or horizontally. Vertical division of labour is based on the establishment of lines of authority and defines the levels that make up the vertical organisational

structure. In addition to establishing authority, vertical division of labour facilities the flow of communication within the organisation.

Horizontal division of labour is based on specialisation of work. The basic assumption underlying horizontal division of labour is that by making each worker's task specialised, more work can be produced with the same effort through increased efficiency and quality.

Advantages of Division of Labour

1. Fewer skills required per person
2. Division of labour (or specialisation) enables a person or an employee performing a task to become highly proficient at it in a relatively short time. As a result, efficiency and productivity increase.
3. Saving the time that is always lost in changing from one job to another.
4. There is less waste of materials in the learning process when division of labour is used.
5. If a job consists of few tasks, a manager can quickly train replacements for personnel who are terminated, transferred, or otherwise absent. The minimum training effort results in lower training cost.
6. The more narrowly defined a job is, the easier it is to develop specialised equipment to assist with that job.
7. The ability to have concurrent operations
8. More conformity in the final product if each piece is always produced by the same person.

Disadvantages of Division of Labour

1. Among the disadvantages of division of labour are boredom, fatigue and dissatisfaction caused by monotonous, repetitive tasks.
2. Overspecialisation could result in workers' having limited knowledge.
3. Different specialities often formulate rules, policies, and procedures that conflict with those of other operational units.
4. Division of labour creates communication barriers. Specialists develop their own language and customs, which can hamper communication across department lines.
5. Specialisation sometimes causes workers to think more in terms of their department or function instead of the company's mission. For example, a computer services department may become so preoccupied with its own activities that it may fail to give managers in other departments the information they need to make timely decisions.
6. Specialisation leads to time-orientation confusion. Production departments, for instance, are commonly short-run oriented; research and development are concerned with the long term. Consequently, production departments typically evaluate their performance in the short run, whereas research and development efforts may go unrecognised for several years.

Division of labour is not more efficient or even desirable in all situations. At least two basic requirements must exist for the successful use of division of labour. The first requirement is a relatively large volume of work. Enough volume must be produced to allow for specialisation and also to keep each employee busy. A second basic requirement is stability in the volume of work, employee attendance, quality of raw materials, product design, and production technology.

Job Design

Job design is concerned with the relationship between workers and the nature and content of jobs, and their task functions. It attempts to meet people's personal and social needs at work through reorganisation or restructuring of work. There are two major reasons for attention to job design:

1. to enhance the personal satisfaction that derive from their work; and
2. to help make the best use of people as a valuable resource of organisation and to help overcome obstacles to their effective performance.

Alternatives to specialisation

To counter the problems associated with specialisation, managers have sought to other approaches to job design that achieve a better balance between organisational demands for efficiency and productivity and individual needs for creativity and autonomy. Some alternative approaches are job rotation, job enlargement and job enrichment.

Job Rotation

Job rotation involves systematically moving employees from one job to another. Job rotation is designed to give an individual broad experience through exposure to many different areas of the organisation. In understudy assignments, coaching, and experience, the trainee generally receives training and development for one particular job. In **job rotation**, the trainee goes from one job to another within the organisation, generally remaining in each from six months to a year. This technique is used frequently by large organisations for training recent university graduates.

One advantage of job rotation is that trainees can see how management principles can be applied in a cross-section of environments. Also, the training is practical and allows the trainee to become familiar with the entire operations of the company. One serious disadvantage of this method is that the trainee is frequently given menial assignments in each job. Another disadvantage is the tendency to leave the trainee in each job longer than necessary. Both of these disadvantages can produce negative attitudes.

Job Enlargement

Job enlargement involves increasing the scope of the job and the range of tasks that the person carries out. It is usually achieved by combining a number of related operations at the same level**. Job enlargement is a horizontal job design; it makes a job structurally bigger**. It lengthens the time cycle of operations and may give the person greater variety.

Job enlargement, however, is not always very popular and may often be resisted by workers. Although it may give the person more to do, it does little to improve intrinsic satisfaction or a sense of achievement. Workers may see job enlargement as simply increasing the number of routine, boring tasks they have to perform. Some workers seem to prefer simple, routine tasks which they can undertake almost automatically, and with little thought or concentration. This enables them to daydream and to socialise with colleagues without affecting their performance.

Job Enrichment

Job enrichment is an extension of the more basic job rotation and job enlargement methods of job design. In job enrichment, there are attempts to enrich the job by incorporating motivating or growth factors such as increased responsibility and involvement, opportunities for advancement and the sense of achievement.

Job enrichment involves vertical job enlargement. It aims to give the person greater autonomy and authority over the planning, execution and control of their own work. It focuses attention on intrinsic satisfaction. Job enrichment increases the complexity of the work and should provide the person with a more meaningful and challenging job. It provides greater opportunities for psychological growth.

Main methods of achieving job enrichment include the following:

- permitting workers greater freedom and control over scheduling and pacing of their work as opposed to machine pacing;
- allowing workers to undertake a full task cycle, build or assemble a complete product or component, or deliver a complete service;
- provide workers with tasks or jobs which challenge their abilities and make fuller use of their training; expertise and skills;
- giving workers greater freedom to work in self-managing teams with greater responsibility for monitoring their own performance and the minimum of direct supervision ; and
- providing workers with the opportunity to have greater direct contact with clients, consumers or users of the product or service.

Grouping Jobs: Departmentation (Departmentalisation)

Departmentation involves grouping jobs into related work units. The work units may be related on the basis of work function, product, customer, location (or geography), process (technique) or time.

Functional Departmentation

The most common base for departmentation, especially among smaller organisations, is by function. Functional departmentation groups together those jobs involving the same or similar activities. (In this context the word function means organisational functions such as finance and production, rather than the basic managerial functions such as planning or controlling.)

This approach has three primary advantages. First, experts in a particular functional area can staff that department. Marketing experts can run the marketing function, for example. Second, supervision is also facilitated because an individual manager needs to be familiar with only a relatively narrow set of skills. And third, co-ordinating activities inside each department is easier than in the other forms of departmentalisation.

On the other hand, as an organisation begins to grow, several disadvantages of this approach may emerge. For one, decision making tends to become slower and more bureaucratic. Employees may also begin to concentrate too narrowly on their own units and lose sight of the total organisational system. Finally, accountability and performance become increasingly difficult to monitor. For example, determining whether a new product fails because of production deficiencies or a poorly received marketing campaign may not be possible.

Product Departmentation

Under **departmentation by product** or service, all the activities needed to produce and market a product or service are usually under a single manager. This system allows employees to identify with a product and thus develop *esprit de corps* (team spirit). It also facilitates managing each

product as a distinct profit centre. Product departmentation provides opportunities for training for executive personnel by letting them experience a broad range of functional activities. Problems can arise if departments become overly competitive to the detriment of the overall organisation. A second potential problem is that facilities and equipment may have to be duplicated. Product 0departmentation adapts best to large, multi-product organisations.

Geographic Departmentation

Geographic departmentation is most likely to occur in organisations that maintain physically dispersed and autonomous operations or offices e.g. Transportation companies, Banks, Police Departments, etc. Departmentation by regions permits the use of local employees and /or salespeople. This can create customer goodwill and an awareness of local feelings and desires. It can provide a high level of service. However, having too many geographic locations can be very costly.

Customer Departmentation

Customer departmentation is based on division by customers served. A common example is an organisation that has one department to handle retail customers, wholesale customers etc. Customer departmentation has the same advantages and disadvantages as product departmentation.

Process Departmentation

Manufacturing firms commonly group activities according to the product's manufacturing processes. Making plywood, for example, involves several sequential processes: poling (removing bark from logs); sawing logs into 8-feet lengths; heating; veneer stripping sheets into 4-feet segments; drying and grading according to quality; gluing plies together to form 4-feet by 8-feet sheets of predetermined thickness; finishing and bundling. The physical layout of a typical manufacturing plant permits raw material to enter the plant, move through the various manufacturing processes, and exit as a finished product.

Establishing Reporting Relationships

Chain of Command

Early writers about the chain of command argued that clear and distinct lines of authority need to be established among all positions in the organisation. The chain of command actually has two components. The first, called unity of command, suggests that each person within an organisation must have a clear reporting relationship to one and only one boss, i.e. an employee must have one and only one immediate manager. The second, called the scalar principle, suggests that a clear and unbroken line of authority must extend from the lowest to the highest position in the organisation, i.e. authority in the organisation flows one link at a time, through the chain of managers ranging from the highest to lowest ranks.

Span of Management

The span of management (also called span of control or span of responsibility) refers to the number of subordinates a manager can effectively manage.

Span of management varies from one situation to another. Upper-level managers usually have fewer immediate subordinates than middle managers do. Similarly, middle managers typically have narrower spans of control than supervisors do. Chief executive officers of medium and large

corporations typically have between four and eight immediate subordinates, for example, while supervisors' spans frequently exceed fifteen persons.

In part, the variations at different organisational levels result from the complexity and/ or diversity of jobs, the need for personal interaction, and leadership philosophies, and a host of other variables.

Wide and Narrow Spans

Another part of establishing reporting relationships is determining how many people will report to each manager, that is determining the span of management. A manager who has a large number of subordinates has a wide span of management. A manager with only a few subordinates has narrow span. A wide span of management means that there are relatively few managerial levels and therefore the organisation is flat. A narrow span of management leads to a tall organisation.

In general, flat organisations tend to be characterised by greater communication between upper and lower-level management, an increased capacity to respond to the environment, and lower total managerial costs than tall organisations.

Limits on the Span of Management

Various writers and theorists have made proposals about the best span of control. V.A. Graicunas, a French management consultant wrote a famous paper on this topic in 1933. Graicunas attempted to demonstrate mathematically the increase in the number of relationships arising from each increase in the number of subordinates reporting to the manager. For example, with one subordinate, the total number of possible relationships was still one but for two subordinates it rose to six relationships. The formula for Graicunas' number of superior-subordinate relationship is given by:

$$R \quad n(2^{n-1} + n - 1)$$

where

R number of relationships

n number of subordinates

In this formula, each additional subordinate creates a considerable number of relationships as shown in Table 2.

Table 2: Possible superior-subordinate relationships

Number of subordinates	Possible number of relationships
2	6
3	18
4	44
5	100
6	222
7	490
8	1080
9	2376
10	5210

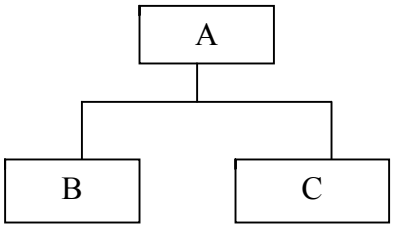
This concept of the span of control is a very theoretical one. It bears no relationship to what happens in practice. Graicunas failed to recognise that not all relationships are of equal importance.

He also did not realise that actual relationships are more potentially limiting than potential relationships.

The value of Graicunas' idea is that it does draw attention to the potentially rapid increase in the complexity of management and supervisory role, when the span of control rises beyond a certain point.

Graicunas noted that groups contain three basic kinds of relationships: single direct (the manager's one-on-one relationship with each subordinate), cross (relationship among subordinates), and group (relationships between groups of subordinates).

V.A. Graicunas' single, group and cross relationships.



Type of Relationship	Number of Relationships
Single Direct: A to B and A to C	2
Group (A and B with C) (A and C with B)	2
Cross: B to C and C to B	2
Total	6

Factors affecting the Span of Management

1. ***Routineness and simplicity of work***: Managers supervising people with simple and repetitive jobs are able to manage more immediate subordinates than are those who supervise people with complex, non repetitive tasks. Tasks are usually more repetitive at lower levels than at upper levels of an organisation. This is a partial explanation of why spans vary at different organisational levels.
2. ***Geographic dispersion of subordinates:*** Normally there is an inverse relationship between a manager' span of control and the geographic dispersion of his subordinates.
3. ***Subordinate training and experience***: The amount of training, experience and ability that subordinates have is directly related to a manager's span of control.
4. ***Use of assistants***: The number of assistants that a manager has is related to span of management. That is, the greater the support given a manager, including the more assistants the manager has to help handle details, the wider the span of management can be.

Distributing Authority

Responsibility, Power and Authority

In the context of organisation **responsibility** can be considered to be the same as accountability. Responsibility is the obligation to make sure that authority is properly used and that duties are

properly carried out. Responsibility carries with it the prerogative to delegate authority and duties, but does not carry the right to avoid accountability. In this sense responsibility flows from the bottom of the organisational structure to the top since each supervisor or manager is accountable to his immediate superior for the proper use of his authority and the proper performance of those duties. In this way the Chief executive carries the full and ultimate responsibility for the effective functioning of the organisation.

Power is the ability to influence, command or apply force. Power is usually derived from the control of resources. The sources of power in organisations are reward, coercive, legitimate, expert and referent. Table 1 shows the basis relating to various sources of power.

Table 1 Sources of Power

Organisational Sources	***Basis***
Reward power	Capacity to provide rewards
Coercive power	Capacity to punish
Legitimate power	Person's position in the organisational hierarchy
Personal Sources	***Basis***
Expert	The skill, expertise, and knowledge an individual possesses
Referent	The personal characteristics of an individual that make other people want to associate with the person

An individual can have power – expert or referent without having formal authority. A manager's authority can diminish by reducing the coercive and reward power in the position.

Authority is the power to assign duties to subordinates and to ensure that they are carried out, and involves the acceptance of accountability for the proper exercise of this authority. It is the legitimate exercise of power. The precise extent of the authority must be clearly defined to the holder and he must act only within those limits. Authority, unlike responsibility, can be delegated and flows downwards (i.e. from top to bottom) through the organisation structure. At all times authority must be commensurate with accountability imposed and all subordinate staff subject to the authority must be made aware of it and of its extent.

Decentralisation and Centralisation

Just as authority can be delegated from one individual to another, organisations also develop patterns of authority across a wide variety of positions and departments. Decentralisation is the process of systematically delegating power and authority throughout the organisation to middle and lower-level managers. Hence, a decentralised organisation is one in which decision-making power and authority are delegated as far down the chain of command as possible. Decentralisation, is just one end of a continuum anchored at the other end by centralisation, the process of systematically retaining power and authority in the hands of higher-level managers. In a centralised organisation decision-making power and authority are retained at the higher levels of management.

LEADING

Leading involves directing and channelling human behaviour toward the accomplishment of objectives.

Leadership

The organisation is made up of groups of people. An essential part of management is co-ordinating the activities of groups and directing the efforts of their members towards the goals and objectives of the organisation. This involves the process of leadership and the choice of an appropriate form of behaviour. The manager must understand the nature of leadership and factors which determine the effectiveness of the leadership relationship.

Leadership is the ability to influence people to willingly follow one's guidance or adhere to one's decisions. Obtaining followers and influencing them in setting and achieving objectives makes a **leader**. Leaders use power in influencing group behaviour. For instance political leaders often use referent power and expert power. Some managers rely on authority, while others use different combinations of power.

The Qualities or Traits Approach to Leadership

1. ***Self-confidence:*** the belief in one's own power to succeed in solving problems and one's own ability to deal effectively with difficult situations.
2. ***Drive:*** the urge and enthusiasm to stimulate action, both personal and in other people.
3. ***Initiative:*** the ability to lead action without waiting to be prompted and to bring fresh thought to old problems.
4. ***Decisiveness:*** the ability to think positively and without vacillation (i.e. to keep changing one's mind) and to act in the same way.
5. ***Willingness to accept responsibility:*** this includes the acceptance of accountability for one's decisions and the consequent acts of oneself and one's staff.
6. ***Ability to delegate:*** this also includes (5)
7. ***Integrity:***; this includes trustworthiness and loyalty to one's organisation and one's staff.
8. ***Judgement:*** the ability to analyse a situation and formulate appropriate action. This includes choosing between different possible courses of action where choices present themselves.
9. ***Adaptability:*** the ability to change one's outlook as circumstances change and to alter one's action in the light of what is necessary in changed circumstances. This is a most valuable characteristic in the modern world which is subject to such rapid change.
10. **Organising ability**;
11. ***Stamina***; the ability to work long and hard without undue strain or stress.
12. ***Emotional maturity***: includes self-discipline and self-control and the ability to analyse a situation without prejudice.
13. ***Human understanding:*** the ability to work with other people with understanding and sympathy; the willingness to listen to staff problems.
14. ***Adequate educational standard:*** this does not mean, necessarily the holding of a university degree, but rather a development of the mind that enables one to think positively and without prejudging a situation, and also to communicate effectively.

Leadership Styles

In the work situation it has become increasingly clear that managers can no longer rely sorely on the use of their positions in the hierarchical structure as a means of exercising the functions of leadership. In order to get the best results from subordinates the manager must also have regard for the need to encourage high morale, a spirit of involvement and co-operation, and a willingness to work. This gives rise to consideration of the style of leadership.

Leadership style is the way in which the functions of leadership are carried out, the way in which the manager typically behaves towards members of the group.

Autocratic Style (Authoritarian or Dictatorial Style)

This is where the focus of power is with the manager, and all interactions within the group move towards the manager. The manager alone exercises decision-making and authority for determining policy procedures for achieving goals, work tasks and relationships, control of rewards or punishments.

Leader

1. The individual is very conscious of his or her position
2. He or she has little trust and faith in members of the group
3. This leader believes pay is a just reward for working and the only reward that will motivate employees.
4. Orders are issued to be carried out, with no questions allowed and no explanations given.

Group members

1. No responsibility is assumed for performance, with people merely doing what they are told.
2. Production is good when the leader is present, but poor in the leader's absence.

Laissez-Faire (Free Reign) Style

A genuine laissez-faire style is where the manager observes that members of the group are working well on their own. The manager consciously makes a decision to pass the focus of power to members, to allow them freedom of action and not to interfere; but is readily available if help is needed. There is often confusion over this style of leadership behaviour. The word "genuine" is emphasised because this is to be contrasted with the manager who could not care, who deliberately keeps away from the trouble spots and does not want to get involved. The manager just lets members of the group get on with the work in hand. Members are left to face decisions which rightly belong with the manager. This is more a non-style of leadership.

Leader

1. He or she has no confidence in his or her leadership ability.
2. This leader does not set goals for the group

Group members

1. Decisions are made by whoever in the group is willing to do it.
2. Productivity generally is low, and work is sloppy
3. Individuals have little interest in their work
4. Morale and teamwork generally are low

Democratic (Participative) Style

The democratic style is where the focus of power is more with the group as a whole and there is greater interaction within the group. The leadership functions are shared with members of the group and the manager is more part of a team. The group members have a greater say in decision-making, determination of policy, implementation of systems and procedures.

Leader

1. Decision-making is shared between the leader and the group.
2. When the leader is required or forced to make a decision, his or her reasoning is explained to the group
3. Criticism and praise are given objectively.

Group members

1. New ideas and change are welcomed.
2. A feeling of responsibility is developed within the group.
3. Quality of work and productivity generally are high.
4. The group generally feels successful.

Delegation

Delegation is the process of allocating tasks to subordinates, giving them adequate authority to carry out those assignments, and making them obligated to complete the tasks satisfactorily.

Steps in the Delegation Process

1. Analyse how you spend your time.
2. Decide which tasks can be assigned.
3. Decide who can handle each task.
4. Delegate the authority.
5. Create an obligation (responsibility).
6. Control the delegation.

Advantages of Delegation

Delegating is important for several reasons.

1. Delegating frees a manager from some time-consuming duties that can be adequately handled by subordinates and lets the manager devote more time to problems requiring his or her full attention.
2. Decisions made by lower-level managers are usually more timely than those that go through several layers of management.
3. Subordinate managers can reach full potential only if given the chance to make decisions and to assume responsibility for them.

Disadvantages of Delegation

1. Managers must be trained to make the decisions that go with delegated authority. Formal training programs can be quite expensive and the expense can more than offset the benefits.

2. Many managers are accustomed to making decisions, and they resist delegating authority to their subordinates. Consequently, they may perform less effectively because they believe that delegation of authority involves loosing control.
3. Administrative costs are incurred because new or altered accounting and performance systems must be developed to provide top management about the effects of their subordinates' decisions. When lower levels of management must have authority top management must have some means of reviewing the use of that authority. Consequently they typically create reporting systems that inform them of the outcomes of the decisions made at lower levels in the organisation.
4. Decentralisation means duplication of functions. Each autonomous unit must be truly self-supporting to be independent. But that involves the potentially high cost of duplication, and some organisations find that the cost of decentralisation outweighs the benefits.

STAFFING

Staffing, usually called employment, involves determining human resource needs and recruiting, selecting, training, and developing human resources. Typically, anticipating human resource needs is the responsibility of line managers. As organisations grow in terms of size and complexity, however, line managers become more dependent on the personnel department to gather information regarding the composition of the work force and the skills of present employees.

Although recruiting is performed primarily by the personnel department, other departments may be involved in providing job descriptions and job specifications to assist the recruiting effort. Also, line managers may be asked to make recruiting visits to University campuses.

In selecting new employees, the personnel department typically screens applicants through interviews, tests, and background investigations, then recommends three or four eligible applicants to the manager or supervisor for a final hiring decision.

Differences Between Line and Staff

The manager in most modern organisations has to deal with people in two types of jobs: line positions and staff positions. The most obvious difference between line and staff is purpose-line managers work directly toward organisational goals, whereas staff managers advise and assist. But other distinctions exist as well. One important difference is authority. Line authority is generally thought of as the formal or legitimate authority created by the organisational hierarchy. Staff authority is less concrete and may take various forms. One form is the authority to advise. In this instance, the line manager can choose whether to seek or avoid input from the staff; even when advice is sought, the line manager might choose to ignore it.

Another form of staff authority is called compulsory advice. In this case the line manager must listen to the advice but can choose to heed it or ignore it. For example, the Pope is expected to listen to the advice of the Sacred College when dealing with church doctrine, but he may follow his own beliefs when making decisions. Perhaps the most important form of staff authority is called functional authority - formal or legitimate authority over activities related to the staff member's specialty. For example, a human resource staff manager may have functional authority when there is a question of discrimination in hiring. Conferring functional authority is probably the most effective way to use staff positions because the organisation is able to take advantage of specialised expertise while maintaining a chain of command.

CONTROLLING

The basic premise of organisations is that all activities will function smoothly; however, the possibility of this being false gives rise to the need for control. **Control** simply means knowing what is actually happening in comparison to preset standards or objectives and then making any necessary corrections. The overriding purpose of all management controls is to alert the manager to a problem or a potential problem before it becomes critical. Control is accomplished by comparing actual performance to predetermined standards or objectives and, then taking action to correct any deviations from the standard. However, control is a sensitive and complex part of the management process.

Controlling is similar to planning. It addresses these basic questions:

- Where are we now?
- Where do we want to be?
- How can we get there from here?

But controlling takes place after the planning is completed and the organisational activities have begun. Whereas most planning occurs before action is taken, most controlling takes place after the initial action has been taken. This does not mean control is practiced only after problems occur; it can be preventive. Control decisions can also affect future planning decisions.

Why Practice Management Control?

Management controls alert the manager to potentially critical problems. At top management levels, a problem occurs when the organisation's goals are not being met. At middle and lower levels, a problem occurs when the objectives for which the manager is responsible are not being met. These may be departmental objectives, production standards, or other performance indicators. All forms of management controls are designed to give the manager information regarding progress. The manager can use this information to:

1. ***Prevent crisis***: If a manager does not know what is going on, it is easy for small, readily solved problems to turn into crisis.
2. ***Standardise outputs***: Problems and services can be standardised in terms of quantity and quality through the use of good controls.
3. ***Appraise employee performance***: Proper controls can provide the manager with objective information about employee performance.
4. ***Update plans***: Even the best plans must be updated as environmental and internal changes occur. Controls allow the manager to compare what is happening with what was planned.
5. ***Protect an organisation's assets***: Controls can protect assets from inefficiency, waste, and pilferage.

Two Concerns of Control

When practicing control, the manager must balance two major concerns: stability and objective realisation. To maintain stability, the manager must be sure the organisation is operating within its established boundaries of constraint. The boundaries of constraint are determined by policies, budgets, ethics, laws, and so on. The second concern, objective realisation, requires constant monitoring to ensure that enough progress is being made toward established objectives.

A manager may become overly worried about one concern at the expense of the other. Most common is a manager who becomes pre-occupied with the stability of the operation and neglects

the goal. This manager is overly concerned with day-to-day rules and policies and forgets about his or her goals. A manager obsessed with the manner or style in which a job is done is an example. On the other hand, a manager may lose sight of stability and have glamorous but short-lived success. A manager who sets production records by omitting safety checks is an example of this behaviour.

The Management Control Process

Three Requirements for Control

The process of control has three basic requirements:

1. establish standards;
2. monitor results and compare to standards; and
3. correct deviations.

The first (setting standards) comes from the planning process, while the latter two (monitoring and correcting) are unique to the control process. All three are essential to having effective control.

Setting Standards **A standard** is a value used as a point of reference for comparing other values. As such, a standard outlines what is expected of the job and/or individual. Standards, when used in management control, often come directly from objectives. In some instances, objectives may be used directly as standards. In other instances, performance indicators may be derived from the objectives. In certain situations, a standard may be determined by methods analysts, industrial engineers, or recognised experts. In any case, standards should be easy to measure and define. The more specific and measurable an objective is, the more likely it can be directly used as a standard. Standards may deal with output per hour, quality level, inventory level, or other indicators of individual and/or organisational performance. The same guidelines for setting objectives should be followed for setting standards.

Monitoring Performance Obviously, the entire control system is no better than the information on which it operates - and much of this information is gathered from the monitoring process. Monitoring is often considered to be synonymous with control. In fact, it is only one part of the total control process. The main purpose of monitoring performance is to gather data and detect problem areas. The type of standards used often dictates the type of checks to be made.

The major problem in monitoring performance is deciding when, where, and how often to inspect or check. Checks must be made often enough to provide needed information. However, some managers become obsessed with the checking process. Monitoring can be expensive if overdone, and it can result in adverse reactions from employees. Timing is equally important. The manager must recognise a problem in time to correct it. For example, inventory control personnel must re-order stock before it has been depleted.

Correcting for Deviations Too often, managers set standards and monitor results but do not follow up with actions. The first two steps are of little value if corrective action is not taken. The action needed may be simply to maintain the *status quo*. Action of this type would depend on standards being met in a satisfactory manner. If standards are not being met, the manager must find the cause of the deviation and correct it. It is not enough simply to treat only the symptoms. This action compares to replacing a car battery when the real problem is a faulty generator. In a short time, the

battery will go dead again. It is also possible that a careful analysis of the deviation will require a change in the standard.

The standard may have been improperly set, or conditions may have changed so as to require a change. Some potential causes of deviations between desired and actual performance are listed as follows:

Potential Causes of Performance Deviations

- Faulty planning
- Lack of communication within the organisation
- Lack of motivation
- Unforeseen forces outside the organisation, such as government regulation or competition

Control Tolerances

Actual performance rarely conforms exactly to standards or plans. A certain amount of variation will normally occur. Therefore, the manager must set limits on the acceptable degree of deviation from the standard. In other words, how much variation from standard is tolerable? For example, how much can expenses be allowed to exceed expectations before the manager becomes alarmed? How much can sales exceed the forecast before some action needs to be taken? The manner in which the manager sets control tolerances depends on the standard and methods being used. If sampling is being used to monitor the activity, statistical control techniques can be used. The tolerance levels may be formalised, or they may merely exist in the mind of the manager. The important point is that the manager must develop some guidelines as to what deviation is acceptable (in control) and what is not acceptable (out of control).

How Much Control?

When deciding how much control should be exercised in an organisation, two major factors must be appraised:

1. economic considerations and
2. behavioural considerations.

Economic Considerations Installing and operating control systems cost money. A good quality control system, for instance, requires additional labour, if nothing else. The equipment costs of sophisticated electronic and mechanical control systems can be very high. Ideally, control systems should be installed as long as they save more than they cost.

The costs of implementing a control system can usually be estimated or calculated much more accurately than the benefits. For example, it is difficult to quantify and measure the true benefits of a quality control system. A good quality control system supposedly increases goodwill; but how does one measure this attribute? The decision is obviously much easier when the costs of not maintaining control are either very high or very low. For example, because the cost of low quality control is so high, airlines go to great expense to ensure high control of quality. Despite the measurement problems, management should regularly undertake such a study to ensure that gross misapplications do not occur.

Behavioural Considerations Managers need to be aware of the possible impact of the control system on employees. Most people do not like to work where they think their every move is being watched or questioned. For example, most salespeople do not like to have to account for every

penny spent on meals and entertainment. At the same time, however, very few people like to work where there is no control: an absence of control creates an environment in which people do not know what is expected of them.

Many managers tend to increase controls whenever things are not going according to plan. This is attempted through the use and enforcement of general and impersonal rules about work procedures. These rigid rules are meant to be guides for the behaviour of the organisation members; they also have the unintended effect of showing minimum acceptable behaviour. In organisations where there is little congruence between individual and organisational objectives or there is not a high acceptance of organisational objectives, the effect is a reduction of performance to the lowest acceptable level - people not highly committed to organisational objectives will perform at the lowest acceptable level. Managers view such behaviour as the result of inadequate control; they therefore respond with closer supervision. This increases the visibility of power, which in turn raises the level of interpersonal tension in the organisation. Raising the tension level brings even closer enforcement of the general and impersonal formal rules. Hence, the cycle repeats itself. The overall effect is increased control, increased interpersonal tension, and a lowering of performance.

One problem in deciding the right degree of control is that different people react differently to similar controls. Research has suggested that reactions differ according to personality and prior experiences. Problems can occur from unforeseen reactions to control because of both compliance and resistance. Problems from compliance arise when people adhere to the prescribed behaviour even when it is inappropriate. The salesperson who will not vary from prescribed procedures in order to satisfy a customer complaint is an example of this. Problems resulting from resistance to controls arise when individuals attempt to pre-empt, work around, or sabotage the controls. Distorting a report and padding the budget are forms of control resistance.

Several things can be done to lessen negative reactions to controls. Most are based on good common sense. However, they require a concerted effort by the manager. The tendency is to take these suggestions for granted. They are:

1. Make sure the standards and associated controls are realistic. Standards set too high or too low turn people off. Make sure the standards are attainable but challenging.
2. Involve subordinates in the control-setting process. Many of the behavioural problems associated with controls result from a lack of understanding of the nature and purpose of the controls. It is natural for people to resist anything new, especially if they do not understand why it is being used. Solicit and listen to suggestions from employees.
3. Use controls only where needed. There is a strong tendency to over control. Periodically evaluate the need for different controls. A good rule of thumb is to evaluate every control at least annually. Things change, which makes certain controls obsolete. Remember, over control can produce negative results.

Where Should Control Reside?

For years, it was believed that control in organisations was a fixed commodity that should rest only in the hands of top management. This viewpoint naturally favoured a highly centralised approach to decision making and controlling. However, as decentralised organisations have become more common, controls have been pushed further and further down the hierarchy. It is now recognised that where the controls reside is an important factor in how much control is desirable.

Evidence favours relatively tight controls as long as they are placed as far down in the organisation as possible. This approach has several advantages:

1. it keeps higher-level managers from getting too involved in details;
2. it shows why the control is necessary; and
3. it elicits commitment from lower-level managers.

When controls are spread through many levels of an organisation, caution must be taken to ensure that there are no disagreements about how the controls are distributed. In other words, managers at every level should clearly understand their authority and responsibility.

Types of Control

There are two categories of control methods: behaviour control and output control. Behaviour (or personal) control is based on direct, personal surveillance. The first-line supervisor who maintains a close personal watch over employees is using behaviour control. Output (or impersonal) control is based on the measurement of outputs. Tracking production records or sales figures are examples of output controls.

Research has shown that these two categories of control are not substitutes for each other in the sense that a manager uses one or the other. The evidence suggests that output control occurs in response to a manager's need to provide an accurate measure of performance. On the other hand, behaviour control is exerted when performance requirements are well known and personal surveillance is needed to promote efficiency and motivation. In most situations, organisations need to use a mix of output and behaviour controls because each serves different organisational needs.

One of the most common' mistakes managers make is to assume a new method or system of control will solve problems. Methods and systems, by themselves, cannot control anything! But together with good administration and intelligent interpretation, appropriate methods and systems do provide control. The most appropriate control system is often worthless if not properly administered. Furthermore, an appropriate control system properly administered can only produce information that then requires intelligent interpretation. Thus, the methods and systems discussed should not be viewed as solutions to control problems; rather, they are aids in the control process. Also, many control tools (such as budgets) are developed as part of the planning process but are also used as part of the control process.

Preliminary, Concurrent, or Post-action Control?

In general, methods for exercising control can be described as either preliminary, concurrent, or post-action. Preliminary control methods, sometimes called steering controls, attempt to prevent a problem from occurring. Requiring prior approval for purchases of all items over a certain monetary value is an example. Concurrent controls, also called screening controls, focus on things that happen as inputs are being transformed into outputs. They are designed to detect a problem as it occurs. Personal observation of customers being serviced is an example of a concurrent control. Post-action control methods are designed to detect existing or potential problems after they occur but before they reach crisis proportions. Written or periodic reports represent post-action control methods. Most controls are based on post-action methods.

Several methods or systems of control commonly used by managers today include:

- Budgetary Control (Flexible (variable) budgets, Zero-Base budgeting
- Financial Controls (balance sheets, income statements, and financial ratios)
- Direct Observation
- Written Reports
- Audits
- Break-Even Charts: A Break-Even Chart graphically depicts the relationship of volume of operations to profits.
- Time-related charts/techniques
- Management by objectives and management information systems

FUNDAMENTALS OF ACCOUNTING AND COSTING

Accounting is the process of ***identifying***, ***measuring***, ***recording***, and ***communicating*** financial information about an organisation's activities to permit informed decisions. Business managers use accounting information to predict the consequences of their future actions and to help decide which actions to take. Business managers also use it to evaluate the performance of employees. Investors use accounting information to evaluate business managers and decide where to invest their money. Governments use it to determine taxes owed by individuals and businesses and to make policy decisions.

The process of recording information and maintaining accounting records is called ***bookkeeping***. Accountants are key participants in the design of information systems that include both accounting and non-accounting systems used by business managers. Accountants also design reports to meet various needs for financial information – including financial statements, tax returns, budgets, and special cost analyses. Finally, accounting includes the design and operation of systems that ensure the reliability of information, including audits of financial information reported to people outside the organisation.

The Demand for Accounting Information

Accounting information is useful because it helps people answer questions and make better decisions. The demand for accounting information about a business comes from both outside and inside. People ***outside*** the business usually want summarised information presented in standardised forms and reported at annual or quarterly intervals. In contrast, people ***inside*** the business want detailed information tailored to the needs of specific decisions and reported at frequent intervals – often weekly or monthly.

Outside Demand

The principal demand for accounting information from outside the business comes from capital market participants, governments, and special-interest groups.

Capital Markets help arrange the investment of billions of dollars each year in business activities. The investors that participate in these markets need accounting information about particular businesses. On the basis of this information, investors decide whether to invest in a business or, if an investment was made in the past, whether to keep the investment or sell it to another investor. Bank loan officers use accounting information as a basis for lending decisions.
Governments need accounting information about businesses to determine taxes owed by businesses, to implement a variety of regulatory objectives, and to formulate economic policy.
Special Interest Groups including labour unions, consumer action groups, customers, competing businesses, financial advisers, and the general public – seek accounting information about businesses as a basis for furthering their own interests.

Inside Demand

The principal demand for accounting information from inside the business comes from ***managers*** – whether they are hired or own their own business - and ***employees***.
Managers need accounting information to control both human and material resources of the business and to direct resources to the most promising products or sub-units of the business.
Employees use accounting information about their employer to help plan their careers.

Meeting the Demand for Accounting Information

The demand for accounting information, whether it comes from outside or inside the business, is met by the same accounting system. The data produced by that system are summarised and reported in different ways to meet different needs for information. Accounting and reporting to satisfy the inside demand for accounting information is called ***management accounting***. Accounting and reporting to satisfy the outside demand for accounting information is called ***financial accounting***. Financial accounting produces financial statements (prepared using generally accepted accounting principles), tax returns (prepared as required by laws and government regulations), and other information products.

Financial Statements summarise the performance and status of a business and are issued at least annually. How much better off is the business at the end of the year than it was at the beginning? What are the economic resources of the business and the claims against those resources? How much cash flowed in and out of the business during the year? These and other questions are answered by financial statements prepared for outsiders.

The Financial Statements

The typical business entity issues four financial statements:

1. The balance sheet
2. The income statement
3. The statement of changes in retained earnings
4. The statement of cash flows

Introduction to the balance sheet

Definition:

A balance sheet is a statement of the total **assets** and **liabilities** of an organisation **at a particular date** - usually the last date of an accounting period. The balance sheet is split into two parts:

1. A statement of **fixed assets**, **current assets** and the **liabilities** (sometimes referred to as "**Net Assets**")

2. A statement showing how the Net Assets have been financed, for example through share capital and retained profits.

The Accounting Equation

The whole of financial accounting is based on the accounting equation. This can be stated to be that for a firm to operate it needs resources, and that these resources have had to be supplied to the firm by someone. The resources possessed by the firm are known as **Assets**, and obviously some of these resources will have been supplied by the owner of the business. The total amount supplied by him is known as **Capital**. If in fact he was the only one who had supplied the assets then the following equation would hold true:

$$\text{Assets} = \text{Capital}$$

On the other hand, some of the assets will normally have been provided by someone other than the owner. The indebtedness of the firm for these resources is known as **Liabilities**. The equation can now be expressed as:

$$\text{Assets} = \text{Capital} + \text{Liabilities}$$

It can be seen that the two sides of the equation will have the same totals. This is because we are dealing with the same thing from two different points of view. It is:

Resources: What they are = Resources: Who supplied them
(Assets) (Capital + Liabilities)

It is a fact that the totals of each side will always equal one another, and that this will always be true no matter how many transactions are entered into. The actual assets, capital and liabilities may change, but the equality of assets with that of the total of capital and liabilities will always hold true.

Assets consist of property of all kinds, such as buildings, machinery, stocks of goods and motor vehicles, also benefits such as debts owing by customers and the amount of money in the bank account.
Liabilities consist of money owing for goods supplied to the firm, and for expenses, also for loans made to the firm.
Capital is often called the owner's equity or net worth.

Long-term and Current
To provide additional information to the user, assets and liabilities are usually classified in the balance sheet as:
Current: those due to be repaid or converted into cash within 12 months of the balance sheet date
Long-term: those due to be repaid or converted into cash more than 12 months after the balance sheet date

Fixed Assets
A further classification other than long-term or current is also used for assets. A "fixed asset" is an asset which is intended to be of a permanent nature and which is used by the business to provide the capability to conduct its trade. Examples of **"tangible fixed assets"** include plant & machinery, land & buildings and motor vehicles. **"Intangible fixed assets"** may include goodwill, patents, trademarks and brands - although they may only be included if they have been "acquired". Investments in other companies which are intended to be held for the long-term can also be shown under the fixed asset heading.

Definition of Capital
As well as borrowing from banks and other sources, all companies receive finance from their owners. This money is generally available for the life of the business and is normally only repaid when the company is "wound up". To distinguish between the liabilities owed to third parties and to the business owners, the latter is referred to as the **"capital"** or **"equity capital"** of the company.

In addition, undistributed profits are re-invested in company assets (such as stocks, equipment and the bank balance). Although these "retained profits" may be available for distribution to shareholders - and may be paid out as dividends as a future date - they are added to the equity capital of the business in arriving at the total **"equity shareholders' funds"**.

At any time, therefore, the capital of a business is equal to the assets (usually cash) received from the shareholders plus any profits made by the company through trading that remain undistributed

The Balance Sheet and the effects of Business Transactions

The accounting equation is expressed in a financial statement called the ***Balance Sheet***. The balance sheet provides information about a business' assets and the claims against those assets by the creditors and owners of the business. It is not the first accounting record to be made, but it is a convenient place to start to consider accounting.

The introduction of capital

On January 1, 2009 K. Ananse started a business and deposited GH¢10,000 in his bank account opened specially for the business. The balance sheet would appear:

K. Ananse Co. Ltd.
Balance Sheet as at 1 January 2009

Assets	GH¢	***Capital and Liabilities***	GH¢
Cash at bank	10,000	Capital	10,000
	10,000		10,000

The purchase of an asset by cheque

On 3 January 2009, K. Ananse buys a building for GH¢4,000. The effect of this transaction is that the cash at the bank is decreased and a new asset, i.e. building, appears.

K. Ananse Co. Ltd.
Balance Sheet as at 3 January 2009

Assets	GH¢	***Capital and Liabilities***	GH¢
Building	4,000	Capital	10,000
Cash at bank	6,000		
	10,000		10,000

The purchase of an asset and the incurring of a liability

On 6 January 2009 K. Ananse buys some goods for GH¢1,000 from K. Kuaso, and agrees to pay them some time within the next two weeks. The effect of this is that a new asset, stock of goods, is acquired, and a liability for the goods is created. A person to whom money is owed for goods is known in accounting language as a **creditor**.

K. Ananse Co. Ltd.
Balance Sheet as at 6 January 2009

Assets	GH¢	***Capital and Liabilities***	GH¢
Building	4,000	Capital	10,000
Stock of goods	1,000	Liabilities	
Cash at bank	6,000	Creditor- K. Kuaso	1,000
	11,000		11,000

Sale of an asset on credit

On 10 January 2008 goods which had cost GH¢500 were sold to A. Ntikuma for the same amount of money to be paid later. The effect is a reduction in the stock of goods and the creation of a new asset. A person who owes the firm money is known in accounting language as a **debtor**. The balance sheet now appears:

K. Ananse Co. Ltd.
Balance Sheet as at 10 January 2009

Assets	GH¢	***Capital and Liabilities***	GH¢
Buildings	4,000	Capital	10,000
Stock of goods	500		
Debtor	500	Liabilities	
Cash at bank	6,000	Creditor- K. Kuaso	1,000
	11,000		11,000

Sale of an asset for immediate payment

On 13 January 2009 goods which had cost K Ananse GH¢200 were sold to Y. Okondor paying for them immediately by cheque. Here one asset, stock of goods, is reduced, while another asset, bank is increased. The balance sheet now appears:

K. Ananse Co. Ltd.
Balance Sheet as at 13 January 2009

Fixed Assets	GH¢	***Capital and Liabilities***	GH¢
Buildings	4,000	Capital	10,000
Current Assets			
Stock of goods	300		
Debtor	500	Liabilities	
Cash at bank	6,200	Creditor- K. Kuaso	1,000
	11,000		11,000

The payment of a liability

On 15 January 2009 K. Ananse pays a cheque for GH¢500 to K. Kuaso in part payment of the amount owing. The asset of bank is therefore reduced, and the liability of the creditor is also reduced. The balance sheet now appears:

K. Ananse Co. Ltd.
Balance Sheet as at 15 January 2009

Fixed Assets	GH¢	***Capital and Liabilities***	GH¢
Buildings	4,000	Capital	10,000
Current Assets			
Stock of goods	300		
Debtor	500	Liabilities	
Cash at bank	5,700	Creditor- K. Kuaso	500
	10,500		10,500

Collection of an asset

A. Ntikuma, who owed K. Ananse GH¢500, makes a part payment of GH¢200 by cheque on 31 January 2009. The effect is to reduce one asset, debtor, and to increase another asset, bank. This results in a balance sheet as follows:

K. Ananse Co. Ltd.
Balance Sheet as at 31 January 2009

Fixed Assets	GH¢	***Capital and Liabilities***	GH¢
Buildings	4,000	Capital	10,000
Current Assets			
Stock of goods	300		
Debtor	300	***Liabilities***	
Cash at bank	5,900	Creditor- K. Kuaso	500
	10,500		10,500

It can be seen that every transaction has affected two items. Sometimes it has changed two assets by reducing one and increasing the other. Other times it has reacted differently. A summary of the effect of transactions upon assets, liabilities and capital is shown below.

Example of transaction			
1	Buy goods on credit	Increase Asset (Stock of goods)	Increase Liability (Creditors)
2	Buy goods by cheque	Increase Asset (Stock of goods)	Decrease Asset (Bank)
3	Pay creditor by cheque	Decrease Asset (Bank)	Decrease liability (Creditors)
4	Owner pays more capital into the bank	Increase Asset (Bank)	Increase Capital
5	Owner takes money out of the business bank for his own use	Decrease Asset (Bank)	Decrease Capital
6	Owner pays creditor from private money outside the firm	Decrease Liability (Creditors)	Increase Capital

Balance Sheet Layout

Assets are held and **liabilities are owed** for varying lengths of time. For example cash and inventories tend to turn over more frequently than land, buildings, and accounts payable turn over more frequently than notes payable. Balance sheets classify assets and liabilities into those of short duration – called current assets and liabilities and those of long duration – called long term (or long-lived) assets and long-term liabilities.

In a typical business, it is reasonable to take one year as the dividing line between current and long-term assets and liabilities. However, if the operating cycle of a business is longer than one year, then accounts receivable, inventories and accounts payable turn over much more slowly than for the typical business. In such cases, it is reasonable to extend the dividing line beyond one year, to equal the length of the operating cycle. In other words, assets and liabilities that are expected to turn over much more within the next operating cycle or next year, whichever is longer, are identified on the balance sheet as current assets and liabilities.

A common structure for classified balance sheet is:

Kofi Enso Nyameye Company Ltd.
Balance Sheet as at 31 December 2008

Assets	$	$	$	Capital and Liabilities	$	$
Fixed Assets				Net worth		
Land		25,000		Common stock	275,000	
Building and Equipment	438,000			Preferred stock	100,000	
Less Depreciation Allowance	82,000	356,000		Retained earnings	25,000	
Total Fixed Assets			381,000	Net worth		400,000
Current Assets				Liabilities		
Cash		10,500		Accounts payable	19,700	
Accounts receivable		18,700		Dividends payable	7,000	
Interest accrued receivable		500		Long-term notes payable	16,000	
Inventories		52,000		Bonds payable	20,000	
Total current assets			81,700	Total liabilities		62,700
Total assets			462,700	Total liabilities and net worth		462,700

or

Kofi Enso Nyameye Company Ltd.
Balance Sheet as at 31 December 2008

Assets	$	$	$
Fixed Assets			
Land		25,000	
Building and Equipment	438,000		
Less Depreciation Allowance	82,000	356,000	
Total Fixed Assets			381,000
Current Assets			
Cash		10,500	
Accounts receivable		18,700	
Interest accrued receivable		500	
Inventories		52,000	
Total current assets			81,700
Total assets			462,700
Capital and Liabilities			
Net worth			
Common stock		275,000	
Preferred stock		100,000	
Retained earnings		25,000	
Net worth			400,000
Liabilities			
Accounts payable		19,700	
Dividends payable		7,000	
Long-term notes payable		16,000	
Bonds payable		20,000	
Total liabilities			62,700
Total liabilities and net worth			462,700

Explanation of some accounting terms:

Account: a place in which all the change in one of the business' assets, liabilities, equity, revenues and expenses are recorded.

Account receivable: money due from an individual or another business as payment for the performance of services or the sale of goods on credit.

Account payable: an obligation arising from the purchase of goods or services on credit.

Accrual: an entry that recognises a revenue or an expense and the associated asset or liability.

Accumulated depreciation: the total amount of depreciation that has been recorded for an asset as of any given time.

Assets: economic resources representing expected future economic benefits obtained or controlled by an entity (e.g. cash, accounts, receivable inventory, land, buildings, and equipment.)

Amortisation: systematic and rational allocation of the acquisition cost of an intangible asset over its useful life.

Common stock: shares in a corporation's equity that usually give stockholders both voting rights and a residual equity interest.

Bond: written promises to repay amounts loaned with interest.

Corporation: a business organisation chartered by the state to conduct business as an "artificial person" and owed by one or more persons called shareholders.

Debenture: an obligation secured by the general credit of the business. An unsecured bond; no assets are specifically pledged to guarantee repayment.

Depreciation: systematic and rational allocation of the cost of property, plant, and equipment (but not land) over their useful lives.

Financing activities: means organisations use to pay for their resources such as land, buildings, and machinery.

Goodwill: an intangible asset arising from attributes such as customer satisfaction, product quality, skilled employees, and business location.

Investing activities: the buying and selling of resources that an organisation controls.

Liabilities: probable future sacrifices of economic benefits (e.g. debts owed to creditors)

Note: a legal document given by a borrower to a lender stating the time of repayment and directly, or indirectly, the amount to be repaid.

Operating activities: use of assets to carry out an organisation's plans in the areas of research, development, purchasing, production, distribution, and marketing.

Preferred stock: shares in a corporation's equity that usually do not give stockholders voting rights but grant specific dividends subject to various guarantees and preferences.

Retained earnings: the component of equity in a corporation representing accumulated profits in excess of losses and payments to owners.

Stockholder (shareholder): the owner (holder) of one or more shares of a corporation's stock; an owner of the corporation.

Long term liabilities: Long term liabilities are those obligations of the business that will require payment beyond the next year or next operating cycle, whichever is longer. Common examples are bonds, notes, and mortgages

Transactions: an exchange between a business and one or more external parties. A measurable internal event such as adjustments for the use of assets.

Depreciation

Since property, plant and equipment are used to produce revenues, a portion of the cost of these assets is recognised as an expense in each period in which the assets are used. Land is an exception,

because it is not consumed in operations. This item, **depreciation**, represents the portion of the cost of tangible operating assets (other than land) recognised as expense for each period. Accumulated depreciation is simply the sum of the depreciation expense amount recognised in all periods for assets on hand.

Considerations for making entries into the Balance Sheet

Assets	***Owners' Equity***	***Liabilities***
Cash	Capital stock	Accounts payable
Accounts receivable	Retained earnings	Notes payable
Inventory	Common stock	Bonds payable
Investments	Preferred stock	Taxes payable
Equipment	Treasury stock	
Land		
Buildings		
Motor vehicles		
Goodwill		

Financial Controls

Budgets are probably the most widely used control devices. A budget is a statement of expected results or requirements expressed in financial or numerical terms. In addition to budgets, many managers use other financial information for control. These include balance sheets, income statements, and financial ratios. Regardless of the type of information used, it is only meaningful when compared with either the historical performance of the organisation or with the performance of other organisations. For example, knowing that a company had net income of ninety billion cedis last year does not reveal much by itself. However, when compared to this year's net income of two hundred and fifty billion cedis or with an industry average of two hundred and fifty billion cedis, much more can be determined about the company's performance.

Income Statements

An income statement, sometimes called a profit and loss statement or a summary of operations, shows how much an organisation made or lost during the fiscal year. Basically, an income statement matches all sources of income (sales, interest earned, etc.) against the costs and outlays incurred to operate (cost of goods, overhead expenses, etc.). An example of an income statement is shown below.

While income statements usually reflect the operations of the entire organisation, it is not uncommon for different departments to operate as independent profit centres and in essence, develop their own income statements. It is also not uncommon for managers to set goals based on forecasted income statements. Like balance sheets, income statement can be used to identify certain trends when analysed over several periods of time.

Unlike the balance sheet, which is a point-in-time description, the income statement is a period of time description. Specifically, it is a report on the profitability of business operations during a specified period of time. Although income statements are usually prepared for the period of one year, they may be prepared quarterly or monthly as well.

K. Ananse Co. Ltd.
Income statement
For the Year Ended December 31, 2008

Revenues	$	$
Sales	505,000	
Interest revenue	3,500	
Total revenues		508,500
Expenses		
Cost of goods sold	290,000	
Selling	28,000	
Administrative	35,000	
Other	12,000	
Total expenses		365,000
Income before taxes		143,500
Taxes for year		71,750
Net (income) profit after taxes for year		71,750

The major categories of an income statement are:
Revenues, which include all sales and interest revenue that the company has received in the past accounting period. Revenue is the value of what is received from goods sold, services rendered, and other financial sources. Note the terms ***revenue*** and ***sales***. Most revenue (money coming in to the firm) comes from sales, but there could be other sources of revenue such as rents received, money paid to the firm for use of its patents, or interest earned. It is important not to confuse the terms revenue and income. Some people use them as if they are synonymous. But a glance at the income statement shows that revenue is at the top of the statement and income is at the bottom. Net income is revenue minus sales returns, costs, expenses and taxes.

Expenses which give a summary of all expenses for the accounting period. Some expense values (for example, income taxes, and cost of goods sold) are itemised in other statements.

Business profitability is measured on the income statement as the difference between the revenue and expense of the period; the detailed revenues and expenses are itemised on the income statement. If revenue exceeds expense, the "bottom line" is called net income (or net profit). If the expense exceeds revenue, the difference is called net loss rather than net income. The income statement uses the equation:

Revenues - Expenses = Profit (or Loss)

Other formulas for the income statement are:

Revenue – Cost of goods sold = Gross margin (gross profit)

Gross margin (gross profit) – Operating expenses = Net income or loss before taxes

Net income before taxes – Taxes = Net income (or net loss)
Note:

Operating expenses are the expenses a business incurs in selling goods or providing services and in managing the business. Operating expenses can be divided into selling expenses and administrative expenses.

Selling expenses are the costs of selling goods or selling and providing services e.g. the costs of selling goods usually include such items as advertising expense and salaries of sales staff.
Administrative expenses are the costs of managing the whole business. Such expenses include salaries paid to managers and members of their staff, depreciation on administrative buildings and equipment, and related amounts for insurance, utilities, property taxes and repairs.

Considerations for making entries into the Income statement

Revenues	Cost of Goods Sold	Expenses	
Sales revenue	Cost of buying goods	Wages	Interest
Rental revenue	Cost of storing goods	Rent	Donations
Commissions revenue		Repairs	Licenses
Royalty revenue		Travel	Fees
		Insurance	Supplies
		Utilities	Advertising
		Entertainment	Taxes
		Storage	

Cost of Goods Sold Statement
Cost of goods sold is the cost to the seller of all goods sold during the accounting period. Cost of goods sold includes the cost of the goods plus the costs of preparing the goods for sale (for example transportation (freight) costs). For a manufacturing business (like Coca-Cola Company, Kumasi), cost of goods sold is the cost of producing the goods (including the cost of bottles, syrup, labour, and bottling plants) plus the costs of preparing the goods for sale. Cost of goods sold is the principal expense for most merchandising and manufacturing businesses. Its size in relationship to sales varies from one business to another.

Cost of goods sold is useful in determining exactly how much it costs to make a particular product over a stated time period usually a year. Note that the total cost of the cost of goods sold statement is entered as an expense item on the income statement. This total is determined using the relations:

Direct materials + direct labour = prime cost
Prime cost + factory expense = cost of goods sold.

Work-in-progress-inventories contain the costs of all inventories currently in production.

Finished-goods-inventories contain completed products that are awaiting sale.

Kweku Ananse Co. Ltd.
Statement of Cost of Goods Sold
For the Year Ended December 31, 2008

Materials	$	$
Inventory January 1, 2008	54,000	
Purchases during year	174,500	
Total	228,500	
Less: Inventory December 31, 2008	50,000	
Cost of materials		178,500
Direct labour		110,000
Prime cost		288,500
Factory expense		7,000
Factory cost		295,500
Less: Increase in finished goods inventory during year		5,500
Cost of goods sold		290,000

Financial Ratio Analysis

Ratio analysis is an examination of financial statements conducted by preparing and evaluating a series of ratios. Ratios (or financial ratios), like other financial analysis data, normally provide meaningful information only when compared with ratios for the same firm (using previous statements) or similar firms. Ratios help analysts by removing most of the effects of size differences. When cedi (dollar/euro, pound sterling) amounts are used directly, size differences between firms may make a meaningful comparison impossible. However, properly constructed financial ratios permit the comparison of firms of quite different size.

Financial ratios can be divided into four basic types: ***liquidity, debt, profitability, and activity***.

Liquidity ratios

Liquidity ratios are used to judge how well an organisation will be able to meet its short-term financial obligations.

Short-Term Liquidity Ratios

Short-term creditors want to know the likelihood that a business will be able to pay its current obligations as they come due. The cash necessary to pay liabilities as they mature will come from the business' available cash and from assets that can be converted into cash readily. Receivables, marketable securities, and inventories can all be converted into cash relatively quickly. Property, plant and equipment and other long-term assets are liquidated much more slowly, and are therefore not likely source of cash for meeting current liabilities. Accordingly, short-term liquidity ratios are based exclusively on current, or short-term assets.

Two ratios are typically used to measure a business' short-term liquidity: current ratio and the quick ratio (also known as acid-test ratio). Both focus on the relationship between current assets and current liabilities. The quick ratio is the more conservative (cautious) measure of liquidity.

Current Ratio

Since a business must meet its current obligations primarily by using its current assets, the current ratio is especially useful to short-term creditors. Current ratio is used to analyse the company's working capital condition. Expressed as an equation,

$$\text{Current ratio} = \frac{\text{Total current assets}}{\text{Total current liabilities}}$$

Example: From the balance sheet above (page 53)
Current assets = \$81,700
Current liabilities = \$26,700

$$\text{Current ratio} = \frac{\$81,700}{\$26,700} = 3.06$$

Since current liabilities are those debts payable in the next year, the current ratio value of 3.06 means that the current assets would cover short-term debts three times. Current ratio values of 2.0 to 3.0 are common. Since some current assets cannot be converted into cash quickly (for example inventory, or prepaid expenses), a current ratio of 2.0 – 3.0 is considered desirable. For comparison purposes, it is necessary to compute the current ratio, and all other ratios for several companies in the same industry.

Acid-test ratio (quick ratio)
Some analysts believe that the current ratio overstate short-term liquidity (i.e. working capital invested in inventory can be converted to cash rapidly). They argue that prepaid expenses (expenses for which payments are made before consumption) often cannot be converted into cash. Further, inventories must be sold and receivables collected from those sales before cash is obtained to pay current liabilities. Both the sale of inventory and collection of receivables can require a lengthy period. Conservative analysts argue that only those current assets that can be turned into cash almost immediately should be used to measure short-term liquidity. The more conservative measure of short-term liquidity is based on Acid-test ratio. Mathematically,

$$\text{Acid test ratio} = \frac{\text{Current assets} - \text{inventories}}{\text{Current liabilities}}$$

Example: From the balance sheet above (page 53)
Current assets = \$81,700
Inventories = \$52,000
Current liabilities = \$26,700

$$\text{Acid test ratio} = \frac{\$(81,700 - 52,000)}{\$26,700} = 1.11$$

An acid test ratio of 1.0 is considered sound. This because there are just enough liquid assets available to satisfy current liabilities. If an organisation can not meet its short-term debt, it often has to go to a high-cost lender for financial assistance.

The Acid test ratio is considered by conservative analysts as being more meaningful for emergency situation when the firm must cover short-term debts using its readily convertible assets.

Creditors typically prefer that both these measures of short-term liquidity be as high as possible. However, because investments in current assets (especially cash and receivables) earn very small

returns compared with the returns on investments in non current assets, management must minimise the proportion of capital invested in current assets if it is to maximise profit.

Debt (or leverage) Ratio

Debt (or leverage) ratios measure the magnitude of owners and creditors' claims on the organisation and indicate the organisation's ability to meet long-term obligations. The **debt-to-equity ratio** and **total debt to total assets ratio** are two common debt ratios. Debt (leverage) ratios refer to the degree to which a firm relies on borrowed funds in its operations. A firm that takes too much debt could have problems repaying its lenders or meeting promises to shareholders. These two groups are generally very interested in a firm's ratios. The ***debt to owners equity measures the degree to which the company is financed by borrowed funds that must be repaid***:

$$\text{Debt to owners' equity ratio} = \frac{\text{Total liabilities}}{\text{Owner's equity}}$$

Example: From the balance sheet (page 53)
Total liabilities =\$62,700
Owners' equity =\$400,000

$$\text{Debt to owners' equity ratio} = \frac{\$62,700}{\$400,000} = 0.16 \text{ or } 16\ \%$$

A ratio of 1 (or 100 per cent) shows that a firm actually has more debt than equity. It is a good bet this firm could be quite risky to both lenders and investors. But remember, it is always important to compare ratios to other firms in the same industry because debt financing is more acceptable in some industries than others. Comparisons with past year's ratios can also identify trends within the firm or industry.

Profitability (or Performance) Ratio

A firm's financial performance is measured by using profitability ratios (also called performance ratios). Profitability ratios measure how effectively the firm is using its various resources to achieve profits. Management's performance is often carefully measured by using profitability ratios. Three of the more important ratios used are earnings per share, income ratio or return on sales and return on equity.

Earnings per share measures the profit a company earns for each share of common stock it has outstanding. Earnings per share measures the overall profit generated for each share in existence over a particular period. This is a crucial ratio for corporations since earnings stimulate growth in the company and pay for shareholders' dividends. Earnings per share is calculated as follows:

$$\text{Earnings per share} = \frac{\text{Net income}}{\text{Number of common shares outstanding}}$$

Earnings per share is an excellent indicator of a firm's current performance and growth on a year-by-year basis.

Another reliable measure of performance is obtained by using a ratio based on the ***income ratio***. Firms use this ratio to see if they are doing as well as their competition in generating income from their sales. Income Ratio is calculated by comparing a company's net income with its total revenue.

$$\text{Income Ratio} = \frac{\text{Net income (or net profit)}}{\text{Total revenue}}$$

Return on equity measures how much was earned for each amount of money invested by owners. It is calculated by comparing a company's net income with its total owner's equity:

$$\text{Return on equity} = \frac{\text{Net income}}{\text{Total owners' equity}}$$

The level of risk involved in the industry and competing firms' return on equity are important in assessing a firm's performance. The higher the risk involved, normally the higher the return on equity expected by investors.

Equity Ratio ***is considered by many analysts to be equal in importance to the current ratio as an indicator of financial stability. It is computed as follows:***

$$\text{Equity ratio} = \frac{\text{Shareholders' equity}}{\text{Total assets}}$$

The answer obtained shows the proportion of the firm owned by shareholders.

Profits help companies to grow. Therefore, these and other profitability ratios are considered vital measurements of company growth and management performance.

Activity (or Operating) Ratio

Activity ratios are used to evaluate how effectively a firm is managing some of its basic operations. Asset turnover, inventory turnover, average collection period, and accounts receivable turnover represent some commonly used activity ratios. Converting the firm's resources to profits is a key function of management. Activity ratios measure the effectiveness of the firm's management in using its available assets. The inventory turnover ratio measures the speed of inventory moving through the firm and its conversion into sales. Inventory sitting idling in a business costs money. The more efficiently the firm manages its inventory, the higher the return. If turnover of inventory to net sales is desired, then Inventory turnover ratio is measured as follows:

$$\text{Net sales to inventory} = \frac{\text{Net sales}}{\text{Average inventory}}$$

Net sales is obtained from the income statement and average inventory is obtained from the cost of goods sold statement.

Example: From the income statement (page 9) and the cost of goods sold statement (page 58):

Net sales = \$505,000
Inventory as at 1 January 2008 = \$54,000
Inventory as at 31 December 2008 = \$50,000
Average inventory = \$(54,000+50,000) ÷ 2 = \$52,000

Therefore

$$\text{Net sales to inventory} = \frac{\$505{,}000}{\$52{,}000} = 9.71$$

This ratio means that the average value of the inventory has been sold 9.71 times during the year.

If inventory turnover is related to cost of goods sold is desired, the ratio is given by

$$\text{Inventory turnover ratio} = \frac{\text{Cost of goods sold}}{\text{Average inventory}}$$

Example: From the cost of goods sold statement (page 58):
Cost of goods sold = \$290,000
Inventory as at 1 January 2008 = \$54,000
Inventory as at 31 December 2008 = \$50,000
Average inventory = \$(54,000+50,000) ÷ 2 = \$52,000

$$\text{Inventory turnover ratio} = \frac{\$290{,}000}{\$52{,}000} = 5.58$$

This ratio means that the average length of time that inventories are held before being sold is 365 days divided by 5.58 (that is 65 days).

A lower than average turnover ratio often indicates obsolete merchandise or poor buying practices. A higher than average ratio may note an understocked condition where sales can be lost because of inadequate stock. An acceptable turnover ratio is often determined on an industry-by-industry basis. Managers need to be aware of proper inventory control to ensure proper performance.

It is important to note that ***financial analysis begins where accounting statements end***.

The Impact of Computer Technology in Accounting
Computers can record and analyse data and provide financial reports. Software can continuously analyse and test accounting systems to be sure they are functioning correctly. Computers can help decision making by providing appropriate information, but they can not make good financial decisions independently. Accounting creativity is still a human trait.

Cost Accounting

Cost accounting is the process of accumulating the information managers need to identify, measure, and control costs. It involves accounting for the costs of products and services and is useful for evaluating managerial performance.

Objectives of Cost Analysis

1. To assist in planning decisions, such as the determination of which products to manufacture, the quantities which should be produced and the selling prices. Since planning is addressed to the future, we are interested in future costs for this purpose. Historical costs are useful only in so far as they are reliable indicators of future costs.
2. To assist in the control of operations by maintaining and improving the efficiency with which resources are employed. Control involves comparing the actual costs of current of current operations against their planned costs. It follows that since actual costs are monetary surrogates of the resources which have been exhausted in current operation, we should be interested in replacing those resources. Hence, for this purpose, we require replacement

costs in line with planned costs by highlighting inefficiencies. It may also lead to a revision of planned costs.

3. To assist in pricing of products or services.
4. To assist in the measurement of reported profits.
5. To assist in the collective bargaining process.

Basic Cost Terms

An **opportunity cost** defined as the benefit that is sacrificed when the choice of one action precludes taking an alternative course of action.

Sunk costs are costs that have been incurred in the past.

A cost that can be traced to a particular department is called a **direct cost** of the department.

A cost that is not directly traceable to a particular department is called an **indirect cost** of the department.

Direct labour is the cost of salaries, wages, and fringe benefits for personnel who work directly on manufactured products.

Indirect labour is the cost of personnel who do not work directly on the product, but whose services are necessary for the manufacturing process.

Allocation of Factory Expense (Overhead)

All costs incurred in the production of an item are accounted for by the cost-accounting system. The cost of goods sold statement is obtained from cost accounting. Cost accounting considers the material, direct labour and factory-expense costs by using cost centres. A cost centre is a term, used for the specific overhead account. For example, a department such as KNUST Mechanical Engineering Department or a machine may be a cost centre. All costs incurred in the department or in the utilisation of the machine are thus collected under one cost-centre title, such as Lathe machine 1.

Since direct materials and labour are assignable to a cost centre, the accountant has only to keep track of these costs. Keeping track of costs is no easy task. Often the cost of the cost accounting system prohibits collection of direct costing data in the detail that the accountant or economist might desire. By far the most burdensome chore of cost accounting is the allocation of factory expense or overhead.

The costs associated with property taxes, service and maintenance departments, personnel, supervision, utilities, etc. must be allocated to the using cost centre. Detailed collection of these data is cost prohibitive and often impossible. Thus allocation schemes are used to distribute the expenses on a reasonable basis. A description of some of the bases is given in Table 1.

Table 1: Factory-expense allocation basis

Overhead category	Allocation basis
Taxes	Space occupied
Air-conditioning, Heat, Light	Space, usage, number of outlets
Power	Space, direct labour hours, direct labour cost, machine hours
Receiving, purchasing	Cost of materials, number of orders, number of items
Personnel, machine shop	Direct labour hours, direct labour cost
Building maintenance	Space occupied, direct labour cost
Software development	Cycle time, throughput
Computers in manufacture	Number of transaction

The most common bases used are direct labour cost and direct labour hours. Most allocation is done using a predetermined factory expense rate. It is given by

$$\text{Overhead rate} = \frac{\text{Estimated overhead costs}}{\text{Estimated basis level}}$$

In general, there has been increase in automation. Therefore the number of direct labour hours used to manufacture a product has decreased substantially. Thus the use of direct labour hours as a basis for overhead allocation is no longer appropriate. This has led to the development of ***indirect cost tracking systems***. These systems are used to directly charge much of the overhead that was historically allocated. Alternatively, different bases have become useful. As shown in Table 1, categories such as computer software used in manufacturing support and control functions - a heavy user of indirect labour – can be allocated using bases such as throughput (production rate) and cycle time (production time per unit).

The estimated overhead cost is the amount allocated to the particular cost centre. For example, if a company has two producing departments, the overhead allocated to each department is used in the above equation when determining the departmental overhead rate.

Problem 1: The following information in Table 2 was obtained from last year's budget for the three machines used to produce bookends. Determine the overhead rate for each machine if the estimated factory expense budget for producing the bookends is $5,000 per machine.

Table 2

Cost source	Allocation basis	Estimated activity level
Machine 1	Labour cost	$10,000
Machine 2	Labour hours	2,000 hours
Machine 3	Material cost	$12,000

Solution:
Using the equation:

$$\text{Overhead rate} = \frac{\text{Estimated overhead costs}}{\text{Estimated basis level}}$$

The rates for the year are:

$$\text{Machine 1} = \frac{\text{Estimated Overhead budget}}{\text{Direct labour cost}} = \frac{\$5{,}000}{\$10{,}000}$$

$$= \$0.50 \text{ per dollar labour}$$

$$\text{Machine 2} = \frac{\text{Estimated Overhead budget}}{\text{Direct labour hours}} = \frac{\$5{,}000}{2{,}000}$$

$$= \$2.50 \text{ per labour hour}$$

$$\text{Machine 3} = \frac{\text{Overhead budget}}{\text{Direct material cost}} = \frac{\$5{,}000}{12{,}000}$$

$$= \$0.42 \text{ per material dollar}$$

Note: Once the product has been manufactured and actual direct labour costs, hours and material costs are computed, each amount of money of direct labour cost spent on machine 1 implies that $0.50 in factory expense will be added to the cost of the product. Similar expense values will be added for machines 2 and 3.

The amount of overhead allocated to an asset is a component of the annual operating cost. Other components of annual operating costs are directly assignable expenses such as insurance, direct labour cost, and some indirect costs.

Determination of Factory Expense and Variance

Once the overhead rates are computed and data collected, it is possible to actually determine the cost of production. This cost may be to operate a department or a machine or to manufacture a product depending upon the manner in which the cost centres are set up.

Assuming that the estimated budget was correct we would expect that the total allocated overhead should equal estimated overhead. Some error in budgeting always exists. There will be some over or under estimation of overhead. The difference between the estimated overhead and actual overhead is known as ***VARIANCE***.

Experience in overhead estimation assists in reducing the variance.

Problem 2 (Continuation from Problem 1):
Determine **the total factory overhead expense** using the following actual data given in Table 3. In addition, calculate the variance for the factory expense.

Table 3: Actual data used for overhead allocation

Cost source	Machine number	Actual cost ($)	Actual hours
Material	1	3,800	
	3	19,550	
Labour	1	2,500	650
	2	3,200	750
	3	2,800	720

Solution: To determine actual production costs, use factory cost relation as given below:
Factory cost = materials cost + labour cost + factory expense (overhead)
To determine factory expense, which is the overhead expense, the rates from the previous example are used as follows:

Machine 1 overhead = (labour cost)(rate) = ($2,500)(0.50)
= $1,250
Machine 2 overhead = (labour hours)(rate) = (750h)($2.50/h)
= $1,875
Machine 3 overhead = (material cost)(rate) = ($19,550)(0.42)
= $8,211

Thus, total factory overhead expense
= Machine 1 overhead + Machine 2 overhead + Machine 3 overhead
= $(1,250 + 1,875 + 8,211)
= **$11,336**.

Factory cost = materials + labour + factory expense (overhead)
= $(3,800 +19,550 + 2,500 + 3,200 + 2,800 +11,336)
= $43,186.

Variance = (Estimated overhead) – (Actual overhead)
= 3 x ($5,000) – $11,336
= **$3,664**

The budget of $15,000 represents a 32.3 % allocation over the actual overhead.

Once estimates of overhead costs are determined, it is possible to make an economic analysis of the present operation versus a proposed anticipated operation.

Problem 3: The ABC Company produces many products, two of which are stereo cabinets and Formica-topped dinette sets (Table 3). A total of $33,000 is allocated to overhead for next year. Determine the overhead rates on the basis of direct labour hours for the two processing and one finishing departments:

a. Individually
b. using an overall basis

Table 3: Overhead allocation for 1 year

Allocation	Stereo processing	Dinette processing	Finishing
Overhead ($)	145,000	145,000	10,000
Direct labour hours	20,000	10,000	6,000

Solution:
a. The rate for each department is computed using

$$\text{Overhead rate} = \frac{\text{Overhead budget}}{\text{Direct labour hours}}$$

Therefore,

$$\text{Stereo Processing} = \frac{\text{Overhead budget}}{\text{Direct labour}} = \frac{145,000}{20,000} = \$7.25 \text{ per hour}$$

$$\text{Dinette Processing} = \frac{\text{Overhead budget}}{\text{Direct labour}} = \frac{145,000}{10,000} = \$14.50 \text{ per hour}$$

$$\text{Finishing} = \frac{\text{Overhead budget}}{\text{Direct labour}} = \frac{10,000}{6,000} = \$1.67 \text{ per hour}$$

b. An overall rate is found by computing

$$\text{Overhead rate} = \frac{\text{Total overhead budget}}{\text{Total direct labour hours}}$$

$$\text{Overhead rate} = \frac{300,000}{36,000} = \$8.33 \text{ per hour}$$

An overall rate is easier to compute and use. However, it cannot account for differences in the nature of the work of departments 'blanketed' under the rate.

ENGINEERING ECONOMY

Basic Concepts and Graphics

The basis of many engineering decisions is an economic one, and engineering economy provides the tools necessary to economically evaluate alternatives. For example, a proposed scheme for the utilization of solar energy must not only be technically feasible, but it must also be economically viable in order to be embraced. The question is how does one determine the economic viability of proposed engineering systems?

Basic Terminology: An ***alternative*** is a stand-alone solution for a given situation. In order to be able to compare different methods for accomplishing a given objective, it is necessary to have an ***evaluation criterion*** that can be used as a basis for judging alternatives. That is, the evaluation criterion is that which is used to answer the question "How will I know which one is best?" In economic analysis, monetary values are generally used as the basis for comparison. Thus, when there are several ways of accomplishing a given objective, the method that has the lowest overall cost is usually selected. However, in most cases the alternatives involve intangible factors, such as the effect of process change on employee morale, which cannot readily be expressed in terms of monetary values. When the alternatives have approximately the same equivalent cost, the non-quantifiable, or intangible, factors may be used as the basis for selecting the best alternative.

Engineering Economy is a decision assistance tool by which one method will be chosen as the most economical one. Engineering economy utilizes a collection of simple mathematical techniques to make rational economic comparisons of various engineering alternatives. These mathematical techniques enable one to account for the time value of money in making comparisons. As a simple example, consider a $1,000 loan. If one borrows $1,000 today and uses this money for ten years, one clearly would be required to repay more than $1,000 at the end of ten years. Similarly, if the loan were repaid in ten equal annual payments, one would expect to pay more than $100 each year. Although the sum of the ten $100 payments is numerically equal to the original $1,000 loan, these ten payments are **equivalent** in economic value only if the **time value of money** is ignored. The **time value of money** means that a monetary amount on one date is not directly comparable with the same monetary amount on another date, even if inflationary influences are negligible. Interest charges (or payments) are a direct consequence of the time value of money. Engineering economy will allow you to take into account the fact that money makes money.

The manifestation of the time value of money is termed interest, which is a measure of the increase between the original sum borrowed or invested and the final amount owed or accrued. Thus if you invested money at some time in the past, the interest would be

Interest = total amount accumulated – original investment

On the other hand, if you borrowed money at some time in the past, the interest would be
Interest = present amount owed – original loan

Interest Calculations

When interest is expressed as a percentage of the original amount per unit time, the result is an *interest rate*. This rate is calculated as follows:

Percent interest rate, i = $\frac{\text{Interest accrued per unit time}}{\text{Original amount}} \times 100\%$

The unit **time period** or **interest period** is typically one year, but shorter interest periods such as one month are also used. The total number of interest periods is denoted by n.

Example 1: Mr. Patrick Cole invested \$100,000 on May 1 and withdrew a total of \$106,000 exactly one year later. Compute

a. the interest gained from the original investment and
b. the interest rate from the investment.

Solution

a.

Interest = total amount accumulated – original investment
= \$106,000 – \$100,000
= **\$6,000**

b.

Percent interest rate, i = $\frac{\$6000 \text{ per year}}{\$100,000} \times 100\%$
= 6% per year

Example 2: Ama Quansimah plans to borrow \$20,000 for 1 year at 15% interest. Compute

a. the interest and
b. the total amount due after 1 year.

Solution

a.

Interest accrued per unit time = $\frac{\text{Original amount} \times \text{Percent interest rate, } i}{100}$

Interest = \$20,000(0.15) = \$3,000

b.

Total amount due is the sum of the principal and interest
Total due = \$20,000 + \$3,000
= \$23,000

Equivalence

The time value of money and interest rate utilized together generate the concept of equivalence, which means that different sums of money at different times can be equal in economic value. For example, if the interest rate is 12% per year, \$100 today (i.e. at present) would be equivalent to \$112 one year from today, since

Amount accrued = 100 + 100(0.12) = 100(1.2)
= \$112

Thus, if someone offered you a gift of \$100 today or \$112 one year from today, it would make no difference which offer you accepted, since in either case you would have \$112 one year from today. The two sums of money are therefore equivalent to each other when the interest rate is 12% per year. At either a higher or a lower interest rate, however, \$100 today is not equivalent to \$112 one year from today. In addition to considering future equivalence, one can apply the same concepts for determining equivalence in previous years. Thus, \$100 now would be equivalent to

$100/1.2 = $89.29 one year ago if the interest rate is 12% per year. From these examples, it should be clear that $89.29 last year, $100 now and $112 one year from now are equivalent when the interest rate is 12% per year.

Simple and Compound Interest
Simple interest is calculated using the principal only. In the calculations of compound interest, the interest for a period is calculated plus the total amount of interest accumulated in previous periods. Thus compound interest means "interest on top of interest" (.i.e., it reflects the effect of the time value of money on the interest too).

Example 3
Felicia Boakye borrows $1000 for 3 years at 14% per year simple interest. How much money will she owe at the end of 3 years?

Solution: The interest for each of the 3 years is
Interest per year = 1000(0.14) = $140
Total interest for 3 years = 1000(3)(0.14) = $420
Finally, the amount due after 3 years = 1000 + 420 = $1420

Example 4
George Goodman borrows $1000 at 14% per year compound interest. Compute the total amount due after a 3-year period.

Solution
The interest and total amount due for each year is computed as follows:
Interest, year 1 = 1000(0.14) = $140
Total amount due after year 1 = 1000 + 140 = $1140
Interest, year 2 = 1140(0.14) = $159.60
Total amount due after year 2 = 1140 + 159.60 = $1299.60
Interest, year 3 = 1299.60(0.14) = $181.94
Total amount due after year 3 = 1299.60 + 181.94 = $1481.54

Symbols and their Meaning
The mathematical relations used in engineering economy employ the following symbols:
P = value or sum of money at a time denoted as the present
F = value or sum of money at some time denoted at some future time
A = a series of consecutive, equal, end-of-period amounts of money
n = number of interest periods; months, years, etc.
i = interest rate per interest period; percent per month, percent per year

Cash Flow Diagrams
Every person has cash receipts (income) and cash disbursements (costs) which occur over a particular time span. These receipts and disbursements in a given time interval are referred to as ***cash flow***, with positive cash flows usually representing receipts and negative cash flows representing disbursements. At any one point in time, the net cash flow would be represented as

Net cash flow = receipts – disbursements

Since cash flow normally takes place at frequent and varying time intervals within an interest period, a simplifying assumption is made that all cash flow occurs at the end of the interest period. This is known as the ***end-of-period*** convention. Thus, when several receipts and disbursements occur within a given interest period, the net cash flow is assumed to occur at the end of the interest period. However, it should be understood that although the monetary amounts of F or A are always considered to occur at the end of the interest period, this does not mean that the end of the period is December 31.

A ***cash-flow diagram*** is simply a graphical representation of cash flows drawn on a time scale. The diagram should represent the statement of the problem and should include what is given and what is to be found. That is, after the cash-flow diagram has been drawn, an outside observer should be able to work the problem by looking at only the diagram. Time 0 is considered to be the present and time 1 is the end of time period 1. (We will assume that the periods are in years).

The time scale of Fig. 1 is set up for 5 years. Since it is assumed that cash flows occur only at the end of the year, we will be concerned only with the times marked 0, 1, 2 5. The direction of the arrows on the cash-flow diagram is important to problem solution. Therefore, a vertical arrow pointing up will indicate a positive cash flow. Conversely, an arrow pointing down will indicate a negative cash flow.

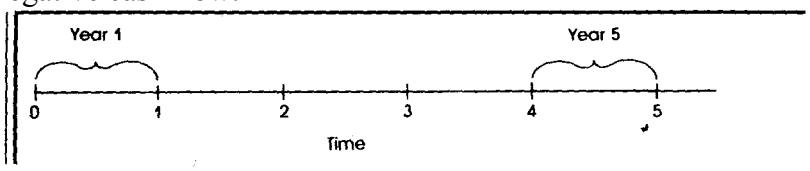


Fig. 1: A typical cash-flow time scale

The cash-flow diagram in Fig. 2 illustrates a receipt (income) at the end of year 1 and a disbursement at the end of year 2. It is important that you thoroughly understand the meaning and construction of the cash-flow diagram, since it is a valuable tool in problem solution. The three examples below illustrate the construction of cash-flow diagrams.

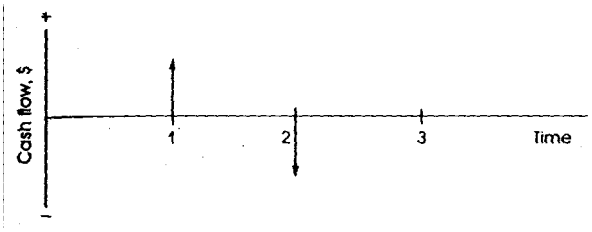


Fig. 2: Example of positive and negative cash flows

Consider the situation presented in the following Example, where P = \$2000 is borrowed and F is to be found after 5 years. Construct the cash-flow diagram for this case, assuming an interest rate of 12% per year.

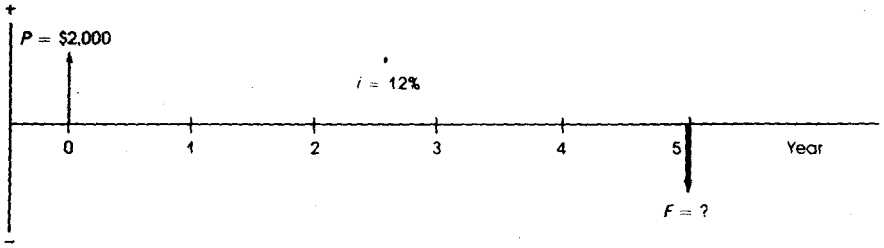


Fig. 3: Cash flow diagram

If you start now and make five deposits of $1000 per year ($A$) in a 17%-per-year account, how much money will be accumulated (and can be withdrawn) immediately after you have made the last deposit? Construct the cash-flow diagram.

Solution: The cash flows are shown in Fig. 4.

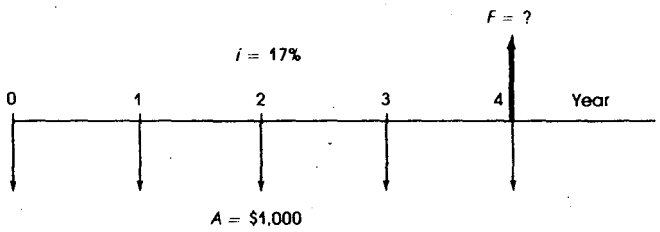


Fig. 4: Cash flow diagram

Assume that you want to deposit an amount P into an account 2 years from now in order to be able to withdraw $400 per year for 5 years starting 3 years from now. Assume that the interest rate is $15\frac{1}{2}$% per year. Construct the cash-flow diagram.

Solution: Fig. 5 presents the cash flows, where P is to be found. Note that the diagram shows what was given and what is to be found and that a P value is not necessarily located at time = 0.

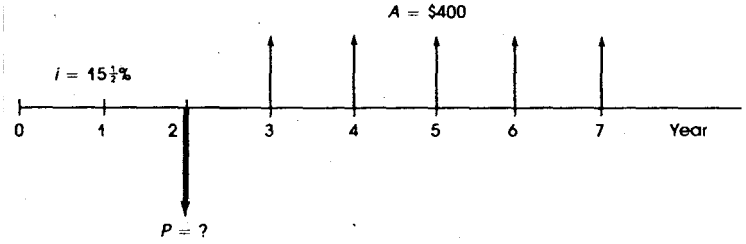


Fig. 5: Cash flow diagram

Calculate the interest and total amount accrued after 1 year if $2000 is invested at an interest rate of 15% per year.

Solution
Interest earned = 2000(0.15) = $300
Total amount accrued = 2000 + 2000(0.15) = 2000(1 + 0.15)
Total amount accrued = $2300

Assignment

Make the calculations necessary to show which of the statements below are true and which are false, if the interest rate is 5% per year:

a. $98 now is equivalent to $105.60 one year from now.
b. $200 one year past is equivalent to $205 now.
c. $3000 now is equivalent to $3150 one year from now.
d. $3000 now is equivalent to $2887.14 one year ago.
e. Interest accumulated in 1 year on an investment of $2000 is $100.

Factors and their use

Standard Factor Notation and Use of Interest Tables

A standard notation has been adopted which represents the, various factors. This standard notation, which also includes the interest rate and the number of periods, will always be of the general form (X/Y, i%, n). The first letter inside the parentheses represents what you want to find, while the second letter Y represents what is given. For example, F/P means "find F when given P." The i is the interest rate in percent and the n represents the number of periods involved. Thus, (F/P. 6%, 20) means obtain the factor which when multiplied by a given P allows you to find the future amount of money F that will be accumulated in 20 periods if the interest rate is 6% per period.

The standard notation is simpler than factor names for identifying factors.. Table 1 shows the standard notation for the formulas. For ready reference the formulas used in computations are collected in Table 2. It is also easy to use the standard notation to remember how the factors may be derived. For example, the A/F factor may be derived by multiplying the P/F and A/P factor formulas.

Table 1: Standard factor notations

Factor name	Standard notation
Single-payment present-worth (SPPWF)	$(P/F, i\%, n)$
Single-payment compound-amount (SPCAF)	$(F/P, i\%, n)$
Uniform-series present-worth (USPWF)	$(P/A, i\%, n)$
Capital-recovery (CRF)	$(A/P, i\%, n)$
Sinking-fund (SFF)	$(A/F, i\%, n)$
Uniform-series compound-amount (USCAF)	$(F/A, i\%, n)$

Table 2: Computations using standard notation

To find	Given	Factor	Equation	Formula
P	F	$(P/F, i\%, n)$	$P = F(P/F, i\%, n)$	$P = F[1/(1+i)^n]$
F	P	$(F/P, i\%, n)$	$F = P(F/P, i\%, n)$	$F = P(1+i)^n$
P	A	$(P/A, i\%, n)$	$P = A(P/A, i\%, n)$	$P = A\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right]$
A	P	$(A/P, i\%, n)$	$A = P(A/P, i\%, n)$	$A = P\left[\frac{i(1+i)^n}{(1+i)^n - 1}\right]$
A	F	$(A/F, i\%, n)$	$A = F(A/F, i\%, n)$	$A = F\left[\frac{i}{(1+i)^n - 1}\right]$
F	A	$(F/A, i\%, n)$	$F = A(F/A, i\%, n)$	$F = A\left[\frac{(1+i)^n - 1}{i}\right]$

In equation form, this is,

$A = F(\not{P}/F, i\%, n)(A/\not{P}, i\%, n)$

$= F(A/F, i\%, n)$

The equivalent of algebraic cancellation of the *P* makes this relation easier to remember. In order to simplify the routine engineering-economy calculations involving the factors above, tables of factor values have been prepared for interest rates from 0.5 to 50% and time periods from 1 to 100. These tables, found in Appendix A and identified as Tables A-1 through A-30, are arranged with various factors across the top and the number of periods *n* down the left and right column. The word "discrete" in the title of each table is printed to emphasize that these tables are for factors which utilize the end-of-period convention and that interest is compounded once each interest period. For a given factor, interest rate, and time, the correct factor value would be found in the respective interest-rate table at the intersection of the given factor and *n*. For example, the value of the factor (*P*/*A*, 5%, 10) is found in the *P*/*A* column of Table A-7 at period 10 as 7.7217. The value 7.7217 could, of course, have been computed using the mathematical expression for the USPWF.

$$(P/A, 5\%, 10) = \frac{(1 + i)^n - 1}{i(1 + i)^n} = \frac{1.05^{10} - 1}{0.05(1.05)^{10}} = 7.7217$$

PROJECT MMANAGMENT

What is a Project?

A project may be defined as a means of moving from a problem to a solution via a series of planned activities. A project has a definite beginning and end. Projects consist of several activities.

Two essential features are present in every project no matter how simple or complicated they are. In the first place, all projects must be *planned* out in advance if they are to be successfully executed. Secondly, the execution of the project must be *controlled* to ensure that the desired results are achieved.

On most projects it is possible to carry out multiple activities simultaneously. Usually it is possible to perform several activities at the same time, however there will be activities which cannot begin until a preceding activity has been completed. Such relationships are referred to as dependencies or precedencies, and when planning a project it is important to establish the order of precedence of dependent activities, and to establish those activities which can be performed in parallel with other activities.

Regardless of the nature or size of your project a successful outcome can only be achieved by using sound project management techniques. The most widely used and popular methods of project management are Gantt Charts, Critical Path Method (CPM) and Programme Evaluation and Review Technique (PERT). However, it is important to remember that projects are carried out by people, and the human aspects of project management are critical for the project success.

Terminology and Definitions

A ***project*** is an interrelated set of activities that has a definite starting and ending point and results in the accomplishment of a unique, often major outcome. "Project management" is, therefore, the planning and control of events that, together, comprise the project. Project management aims to ensure the effective use of resources and delivery of the project objectives on time and within cost constraints.

An ***activity*** or task is the smallest unit of work effort within the project and consumes both time and resources which are under the control of the project manager. A project is a sequence of activities that has a definite start and finish, an identifiable goal and an integrated system of complex but interdependent relationships.

A ***schedule*** allocates resources to accomplish the activities within a timeframe. The schedule sets priorities, start times and finish times.

Project management is: the adept use of techniques and skills (hard and soft) in planning and controlling tasks and resources needed for the project, from both inside and outside of organisation, to achieve results.

The purpose of project management is to achieve successful project completion with the resources available. A successful project is one which:

- has been finished on time
- is within its cost budget
- performs to a technical/performance standard which satisfies the end user.

The Attributes of Successful Project Management

The effectiveness of project management is critical in assuring the success of any substantial undertaking. Areas of responsibility for the project manager include planning, control and implementation. A project should be initiated with a feasibility study, where a clear definition of the goals and ultimate benefits need to be established. Senior managers' support for projects is important so as to ensure authority and direction throughout the project's progress and, also to ensure that the goals of the organization are effectively achieved within this process. The particular form of support given can influence the degree of resistance the project encounters.

Knowledge, skills, goals and personalities are all factors that need to be considered within project management. The project manager and his/her team should collectively possess the necessary and requisite interpersonal and technical skills to facilitate control over the various activities within the project.

The stages of implementation must be articulated at the project planning phase. Disaggregating the stages at its early point assists in the successful development of the project by providing a number of milestones that need to be accomplished for completion. In addition to planning, the control of the evolving project is also prerequisite to success. Control requires adequate monitoring and feedback mechanisms by which senior and project managers can compare progress against initial projections at each stage of the project. Monitoring and feedback also enables the project manager to anticipate problems (e.g.: the knock-on effects of late start or finish times) and therefore take pre-emptive corrective measures for the benefit of the project overall.

Projects normally involve the introduction of a new system of some kind and, in almost all cases, new methods and ways of doing things. This impacts upon the work of others: the "users". User consultation is an important factor in the success of projects and, indeed, the degree of user involvement can influence the extent of support for the project or its implementation plan. A essential quality of the project manager is that of being a good communicator, not just within the project team itself, but with the rest of the organization and outside bodies as well (the users may be internal or external).

Features of projects

a. Projects are often carried out by a team of people who have been assembled for that specific purpose. The activities of this team may be co-ordinated by a project manager.
b. Project teams may consist of people from different backgrounds and different parts of the organisation. In some cases project teams may consist of people from different organisations.
c. Project teams may be inter-disciplinary groups and are likely to lie outside the normal organisation hierarchies.
d. The project team will be responsible for delivery of the project end product to some sponsor within or outside the organisation. The full benefit of any project will not become available until the project as been completed.

In recent years more and more activities have been tackled on a project basis. Project teams and a project management approach have become common in most organisations. The basic approaches to project management remain the same regardless of the type of project being considered. You may find it useful to consider projects in relation to a number of major classifications:

a. Engineering and construction

The projects are concerned with producing a clear physical output, such as roads, bridges or buildings. The requirements of a project team are well defined in terms of skills and background, as are the main procedures that have to be undergone. Most of the problems which may confront the project team are likely to have occurred before and therefore their solution may be based upon past experiences.

b. Introduction of new systems

These projects would include computerisation projects and the introduction of new systems and procedures including financial systems. The nature and constitution of a project team may vary with the subject of the project, as different skills may be required and different end-users may be involved. Major projects involving a systems analysis approach may incorporate clearly defined procedures within an organisation.

c. Responding to deadlines and change

An example of responding to a deadline is the preparation of an annual report by a specified date. An increasing number of projects are concerned with designing organisational or environmental changes, involving developing new products and services.

Activity

Consider this last category. Can you think of any examples where your organisation needs to respond to change and a project management approach may be valid?

Responsibilities of the Project Manager

1. To plan thoroughly all aspects of the project, soliciting the active involvement of all functional areas involved, in order to obtain and maintain a realistic plan that satisfies their commitment for performance.
2. To control the organization of manpower needed by the project.
3. To control the basic technical definition of the project, ensuring that "technical" versus "cost" trade-offs determine the specific areas where optimisation is necessary.
4. To lead the people and organizations assigned to the project at any given point in time. Strong positive leadership must be exercised in order to keep the many disparate elements moving in the same direction in a co-operative.
5. To monitor performance, costs and efficiency of all elements of the project and the project as a whole, exercising judgement and leadership in determining the causes of problems and facilitating solutions.
6. To complete the project on schedule and within costs, these being the overall standard by which performance of the project manager is evaluated.

Why do projects go wrong?

There can be many reasons why projects go wrong. The most common reasons are as follows:

a) Project goals are not clearly defined
b) There can be constraints on the completion of projects arising from the different objectives of:
   - Short time scale
   - Resource availability
   - Quality factors
   - Human factors

Problems with Project Goals

- The project sponsor or client has an inadequate idea of what the project is about at the start.
- There may be a failure of communication between the client and the project manager. The may be due to a lack of technical knowledge on the part of the client or an overuse of jargon by the project manager.
- Specifications may be subject to constant change. This may be due to problems with individual clients, decision making processes at the client end, or environmental changes. For example the government may change the basic "rules of the game" before the completion of the project.
- The project goals may be unrealistic and unachievable, and it may be that this is only realised once the project is under way.
- The client may become carried away with the idea of the project and may be unable to see clearly what can be achieved.
- Projects may be highly complex and may have a number of objectives that actually contradict each other.

There are perhaps two stages which can help in ensuring that goals are properly defined and achievable:

a) Ensuring that the client specification is clear and understandable. To do this you must first of all establish the objectives of the project. It would help to ask the following questions:
   - What is it that the organisation is setting out to achieve or is being asked to achieve?
   - Will the suggested project fulfil these objectives?
   - Have all the alternatives been considered and is the chosen option the best one available?
   - Have the full effects of the project, both inside and outside the organisation, been considered?

b) Preparation of a Project overvies (Project brief). The brief should take the objectives set out in the previous exercise and translate them into targets and goals. Any key constraints should also be identified and stated at this stage. This brief should be agreed by the sponsor/client and communicated to the project manager. Any ambiguities or queries should be sorted out as soon as possible.

A good way forward would be through the establishment of **success criteria** for the project. If you want the project to succeed (and who doesn't?), then you have to know when you have succeeded.

Success criteria can be described as being **hard** or **soft**:

- **Hard criteria** are often the most obvious criteria that are tangible and measurable and can be expressed in quantitative terms. They tend to pose the question "what?", that is "what should be achieved?"
- **Soft criteria** are usually less obvious, but not necessarily less important. They are often intangible and qualitative. Consequently they may be difficult to measure. They would tend to ask the question "how?"

a) Hard criteria

- Performance specifications: these may be set out in terms of the ability to deal with certain demands. For example, this could be throughput of traffic, number of patients, volume of transactions processed or the number of enquiries dealt with.

- Specific quality standards: this could relate to technical standards and tolerance, or may be the achievement of a favourable report from an outside inspection agency.
- Meeting deadlines: this is probably the most obvious one of them all where projects need to be completed within a given time scale. For example, a new system may need to be implemented ready for the start of the financial year, or a new development may have to meet time requirements as laid down in contract specifications.
- Cost of budget constraints: an important criterion may be to complete the project within a cost limit or budget which has been determined. Additionally there may be requirements in terms of the ongoing cost of the completed project. For example, a new system may be required to make savings for the organisation on a continuing basis.
- Resource constraints: there may be other resource constraints such as making use of existing premises or labour force.

As you can see, the above criteria are relatively easy to establish and should also be quite easy to specify in a project brief.

b) Soft criteria

- Demonstrative co-operation: this would be about showing that the project team could work together effectively and without a degree of conflict. It could be an important consideration to develop and implement solutions for the organisation which have a wide element of consensus and stem from a co-operative attitude.
- Presenting a positive image: this may also be important but obviously can be difficult or impossible to quantify.
- Achieving a total quality approach: this would be more about the adoption of a philosophy of continuous improvement than the achievement of specific performance targets on quality.
- Gaining total project commitment: this is again about how the project is managed and the attitude of the project team to it.
- Ensuring that ethical standards are maintained: it can be very important to ensure that no corners have been cut that should not have been and that professional standards of ethics have not been breached.
- Showing an appreciation of risk: this would ensure that no unacceptable risks were taken in the pursuit of other project objectives. Again this is about how the project is developed rather then the end product itself.

Constraints on the completion of projects

a) **Time**

Our definition of a project stated that it was an activity which had a defined beginning and ending point. Most projects will be close-ended in terms of there being a requirement for completion by a certain point in time. This point may be the result of an external factor such as new legislation, or may be derived from organisational requirements. It may also be partly determined by other constraints. There is likely to be some relationship between the time taken for a project and its cost. A trade-off between the two constraining factors may then be necessary.

b) **Resource Availability**

There is likely to be a budget for the project and this will clearly be a major constraint. Cost constraints may be set in a number of ways, for example as an overall cash limit or as a detailed budget broken down over a number of expenditure headings. Labour resources in particular may be a limiting factor on the completion of the project. In the short run it is likely that labour will be fixed in supply. Whilst the overall resource available may in theory be sufficient to complete the project, there may be difficulties arising out of the way in which the project has been scheduled. That is, there may be a number of activities scheduled to take place at the same time and this may not be possible given the amount of resources available.

c) **Quality factors**

Whether the project delivers the goods to the right quality.

There are techniques which can be used to overcome the problems referred to above. These include:

- ***Budgeting***, and the corresponding control of the project budget through budgetary control procedures.
- ***Project planning*** and ***control*** techniques such as Gantt charts and network analysis.

An important point to note at this stage is how the various constraints on project completion are likely to be interlinked with each other. For example, problems with time constraints or resource constraints may be overcome by spending more through working overtime, employing more people or purchasing better machines. Budget problems may have a knock-on effect on the achievement of deadlines.

It is important to remember that while project management techniques are important, they tend to understate the importance of the key resource: people. In a fact changing environment where tasks are often difficult, controversial with uncertain outcomes, "people management" skills are called for.

THE PROCESS OF PROJECT MANAGEMENT

Project Planning

A major decision at the outset of any project is to decide upon the organization and composition of the project team. In so doing, it is worth remembering that many members will have dual responsibilities of involvement in the project in addition to a commitment to other projects or management of a functional area on a day-to-day basis. It is at this stage that a project manager should be appointed and responsibilities made explicit for all members of the team.

The selection of the team will be dependent upon the skill requirements of the project, and upon the matching of those skills to those possessed by individual members of the team. There may be a conflict here with hierarchical status.

The project management team will, therefore, begin its task in advance of project proper so that a plan can be developed. An important first step is to set the objectives and then define the project, breaking it down into a set of activities and related costs. It is probably too early to determine exact resource implications at this stage, but expected requirements for people, supplies and equipment should at least be estimated during the planning stage.

Project Scheduling

This phase is primarily concerned with attaching a timescale and sequence to the activities to be conducted within the project. Materials and people needed at each stage of the project are determined and the time each is to take will be set.

A popular and easy to use technique for scheduling is the use of Gantt charts. Gantt charts reflect time estimates and can be easily understood. Horizontal bars are drawn against a time scale for each project activity, the length of which represent the time taken to complete. Letters or symbols can also be added to the left of each bar to show which other activities need to be completed before that one can begin.

Gantt Charts

A Gantt Chart is a simple technique that can be used to attach a time scale and sequence to a project. A Gantt Chart is a form of horizontal bar chart and horizontal bars are drawn against a time scale for each project activity, the length of which represents the time taken to complete. To construct a Gantt Chart the following steps are necessary:

1. Use the horizontal axis to represent time
2. Use the vertical axis to represent activities
3. Represent each activity by a horizontal bar of appropriate length

Take activity procedures into account by starting each activity bar to an appropriate point along the time axis after its preceding activities. Normally the start point for an activity is the earliest time that it could start after its preceding activities had finished.

It is possible to enhance the Gantt Chart in several ways. For instance the number of staff required to do a task can be entered into the bar on the diagram.

Gantt charts, also commonly known as milestone plans, are a low cost means of assisting the project manager at the initial stages of scheduling. They ensure that:

1. all activities are planned for,
2. the sequence of activities is accounted for,
3. the activity time estimates are recorded; and
4. the overall project time is recorded.

They are therefore a simple, rough and ready means of planning a project and assessing progress and are sufficient for most simple projects.

However, where projects become complex, it becomes difficult to see relationships between activities by using a Gantt Chart. For more complex projects Network Analysis techniques are used.

Gantt charts also provide a summary of the project as a whole and can be used as a rough and ready means of assessing progress at the project control phase. At any date, the project manager can draw a dateline through the Gantt chart and see which activities are on-time, which are behind schedule and generally record project status against plan.

Gantt charts, named after Henry L. Gantt, one of the pioneers of scientific management, are a useful means of representing a schedule of activities comprising a project and enable the operations manager to know exactly what activities should be performed at a given time and, more importantly, to monitor daily progress of a project so that corrective action may be taken when necessary.

To construct a Gantt chart, the various activities are listed on a vertical axis and the horizontal axis is used to represent time. Activity precedencies are taken into account by starting a horizontal bar to represent the next activity at an appropriate point after its preceding activities, i.e. those activities which must take place before the next activity can start, have taken place. Normally this would be at the earliest time that it could start after its preceding activities had finished.

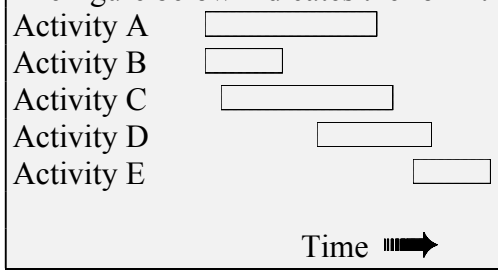
Example:

Suppose a project comprises five activities: A,B,C,D, and E. A and B have no preceding activities, but activity C requires that activity B must be completed before it can begin. Activity D cannot start until both activities A and B are complete. Activity E requires activities A and C to be completed before it can start. If the activity times are A: 9 days; B: 3 days; C: 9 days; D: 5 days; and E: 4 days,

i) determine the shortest time necessary to complete this project.
ii) identify those activities which are critical in terms of completing the project in the shortest possible time.

Solution:

The figure below indicates the form that a Gantt chart would take from the above information.



The above chart shows that activities A and B have no preceding activities and so can start right away. Activity C requires Activity B to have been completed before it can begin. The chart is then completed using such precedence relationships as listed in the question, with each horizontal bar being proportional in length to the activity time that it represents.

Looking at the chart it is apparent that the project ends when activity E has been completed. Working back in time from activity E the "steps" which are crucial or critical in order to ensure that the project duration does not extend beyond the planned length are: E,C and B. The shortest time in which the project could be completed from the given information is therefore 16 days. The set of activities B, C and E which together determine the project duration are referred to as the **critical path** through the chart.

Those activities forming the critical path can be highlighted on the Gantt chart to help the operations manager to give priority to them if lack of resources mean that such decisions have to be made.

It is important to realize though that activities not on the critical path can become so if they are allowed to drift too far. How far could activities A and D drift before they affected the duration of the project?

Provided the project is not too complex in its activity relationships or simply too big to be mapped on reasonably sized graph paper, Gantt charts can be very useful tools for the project manager and are graphically superior to the network analysis methods of CPM and PERT. They allow the critical activities to be found, i.e. those activities which must be performed on time if the project duration is not to increase, and any "slack" or "float" in the sequence of activities can easily be shown.

List of References

1. Daft, R.L. (2003). Management, 6th Edition, Thomson South-Western, Ohio.
2. Robbins, S.P. and Coulter, M. (1999). Management, 6th Edition, Prentice Hall Inc., New Jersey.
3. Griffin, R.W. (1999). *Management*, 6th Edition, Houghton Mifflin Company, Boston.
4. Rue, L.W. and Byars, L.L. (1995). Management, Skills and Application, 7th Edition, Irwin, Chicago.
5. Kinard, J. (1988). *Management*. D.C. Heath, New York.
6. Sangster, A. and Wood, F. (2008). Business Accounting: Volume 1, 11th Edition, FT Prentice Hall,
7. Blank, L. and and Tarquin, A. (2002). Engineering Economy, 5th Edition, McGraw-Hill Inc., New York.
8. Clausing, A.M. (2002). Selected Notes on Engineering Economics, Available online: http://mechse.illinois.edu/media/uploads/course_websites/engeconnotes.20070314.45f80866b903c4.34988219.pdf
9. Field, M. and Keller, L. (2004). Project Management, The Open University, London.